MULTI-STAGE KNEE OSTEOARTHRITIS CLASSIFICATION USING DEEP LEARNING ON DXA IMAGES

A PROJECT REPORT

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KONGU ENGINEERING COLLEGE

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This is to certify that the project report entitled **"MULTI-STAGE KNEE OSTEOARTHRITIS CLASSIFICATION USING DEEP LEARNING ON DXA IMAGES"** is the bonafide record of project work done by **PRAGATHESHWARAN.R (25MCR083), SWETHA.S (25MCR118),VIDHYA.J (25MCR124)** in partial fulfillment of the requirements for the award of the Degree of Master of Computer Applications of Anna University, Chennai during the year 2025-2026.

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Submitted for the end semester viva-voice examination held on____________________

INTERNAL EXAMINER **EXTERNAL EXAMINER**

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We affirm that the project report entitled **“MULTI-STAGE KNEE OSTEOARTHRITIS CLASSIFICATION USING DEEP LEARNING ON DXA IMAGES”** being submitted in partial fulfillment of the requirements for the award of Master of Computer Applications is the original work carried out by us. It has not formed the part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

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ABSTRACT

Knee Osteoarthritis (KOA) is one of the most common degenerative joint diseases, affecting millions of people worldwide and often leading to pain, stiffness, and reduced mobility. Early and accurate diagnosis is essential for effective treatment and disease management. Traditional manual assessment of knee X-ray images is time-consuming and subject to inter observer variability, motivating the need for automated and reliable computer-aided diagnostic systems.

This study proposes a hybrid deep learning framework for automated classification of Knee Osteoarthritis severity using radiographic images. The proposed model combines the strengths of EfficientNetV2-S and ConvNeXt-Tiny architectures to capture both local and global features from knee X-rays. An attention-based fusion mechanism is integrated to enhance feature representation and improve classification performance. To address class imbalance, a Weighted Random Sampler technique is employed during training, while data augmentation and normalization techniques improve model generalization.

The model is trained using the AdamW optimizer with learning rate scheduling and fine tuning strategies. Cross Entropy Loss with label smoothing is used to improve robustness and reduce overconfidence in predictions. Performance evaluation is conducted using classification reports, confusion matrices, ROC curves, accuracy, precision, recall, F1-score, and AUC metrics. Additionally, Grad-CAM visualization and attention heatmaps are incorporated to provide interpretability by highlighting the regions influencing model predictions.

Experimental results demonstrate that the proposed hybrid model achieves high classification accuracy and improved robustness compared to conventional single-network approaches. The integration of attention mechanisms and interpretability techniques enhances both diagnostic reliability and transparency. The proposed system can support radiologists and healthcare professionals in the early detection and grading of Knee Osteoarthritis, contributing to more efficient and accurate clinical decision-making.

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LIST OF ABBREVIATIONS

KOA	Knee Osteoarthritis
KL	Kellgren–Lawrence
CNN	Convolutional Neural Network
DL	Deep Learning
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
Grad-CAM	Gradient-weighted Class Activation Mapping
PIM	Plug-in Module
MRI	Magnetic Resonance Imaging
ROC	Receiver Operating Characteristic
AUC	Area Under Curve
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
F1-Score	Harmonic Mean of Precision and Recall
SGD	Stochastic Gradient Descent
GPU	Graphics Processing Unit
ReLU	Rectified Linear Unit
CNNs	Convolutional Neural Networks

EfficientNet	Efficient Neural Network
OAI	Osteo arthritis Initiative
ConvNext	Convolutional Neural Network
ResNet	Residual Network
DenseNet	Densely Connected Convolutional Network
Adam	Adaptive Moment Estimation
AdamW	Adam with Weight Decay
TPR	True Positive Rate
FPR	False Positive Rate

CHAPTER-1

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

Knee Osteoarthritis (KOA) is one of the most common degenerative joint disorders affecting millions of people worldwide, especially the elderly population. It leads to the gradual breakdown of cartilage in the knee joint, resulting in pain, stiffness, reduced mobility, and decreased quality of life. Early detection and accurate assessment of KOA severity are essential for effective treatment planning and slowing disease progression.

Traditionally, KOA is diagnosed and graded using the Kellgren–Lawrence (KL) gradingsystem**,** which relies on visual inspection of X-ray images by medical experts. However, this method is highly subjective and often leads to inconsistencies between clinicians. The challenge becomes even greater in early-stage detection where the differences in X-ray images are very subtle and difficult to interpret accurately.

With the rapid advancement of artificial intelligence, particularly in deep learning and computer vision, automated systems have been developed to assist in medical image analysis. Convolutional Neural Networks (CNNs) and advanced architectures such as ResNet, DenseNet, and EfficientNet have shown significant success in image classification tasks. These models can learn complex patterns from large datasets and provide more consistent and objective results compared to manual diagnosis.

Despite these advancements, existing models still face challenges such as low accuracy in early-stage detection, class imbalance, limited dataset diversity, and lack of interpretability. To address these issues, recent approaches incorporate Plug-in Modules **(**PIMs) and ensemble techniques to enhance feature extraction and improve classification performance. Additionally, explainable AI techniques like Grad-CAM are used to increase transparency and build trust among clinicians.Therefore, this study focuses on developing a deep learning-based system for accurate and automated classification of KOA severity using knee X-ray images. By leveraging advanced models and improved methodologies, the study aims to provide a reliable, efficient, and clinically useful tool for early diagnosis and better patient care.

1.2 PROBLEM STATEMENT

The need for accurately classifying the severity of Knee Osteoarthritis (KOA) using X-ray images remains a significant challenge in the medical field. The widely used Kellgren–Lawrence (KL) grading system is often subjective and prone to variation among clinicians, especially in the early stages where visual differences in X-rays are subtle and difficult to interpret. Existing deep learning models also struggle to achieve high accuracy in detecting these early stages, limiting their effectiveness in real-world clinical applications. Additionally, issues such as limited and non-diverse datasets, class imbalance, and insufficient feature extraction further reduce model performance and reliability. Therefore, there is a need to develop an accurate, reliable, and automated system that can classify KOA severity across all grades. This system should leverage advanced deep learning techniques, such as convolutional neural networks enhanced with Plug-in Modules (PIMs), to capture fine-grained image details, improve generalization using large datasets, and support faster, consistent, and practical clinical diagnosis.

1.3 OBJECTIVES

- To develop an automated deep learning system for detecting and classifying Knee Osteoarthritis (KOA) from X-ray images.
- To classify KOA severity accurately into five grades based on the Kellgren-Lawrence (KL) grading system.
- To improve early-stage osteoarthritis detection, especially for Osteopenia, Osteoporois cases where visual differences are subtle.
- To use advanced deep learning models such as EfficientNetB3, DenseNet121, and ResNet50V2 for feature extraction and classification.
- To enhance classification accuracy using Plug-in Modules (PIMs) and transfer learning techniques.
- To reduce manual errors and inconsistencies in traditional clinical diagnosis methods.
- To improve model generalization and reliability using large-scale training and testing datasets.
- To apply preprocessing and data augmentation techniques for better model performance and robustness.

1.4 SCOPE AND LIMITATIONS

1.4.1 SCOPE

This study focuses on developing an automated, accurate, and reliable system for detecting and classifying the severity of knee osteoarthritis across all stages using medical imaging. This project aims to replace or assist traditional manual diagnosis methods, such as the Kellgren-Lawrence grading system, which are often subjective and inconsistent. By leveraging advanced deep learning models like CNNs, EfficientNet, DenseNet, and ResNet along with Plug-in Modules (PIMs), the system enhances feature extraction and improves classification performance, especially in early-stage detection where visual differences are minimal. The scope also includes preprocessing and augmenting large-scale X-ray datasets, training and testing models for better generalization, and integrating explainable AI techniques to increase clinical trust. Ultimately, the project is designed to support healthcare professionals by providing faster, consistent, and objective diagnostic results, with potential for real-world clinical deployment and future expansion by incorporating multi-modal data and improved model interpretability.

1.4.2 LIMITATIONS

- The model is trained on a specific dataset (~5000 images), which may not represent all patient populations, imaging conditions, or hospital variations—reducing generalization.Detecting osteoarthritis is difficult because differences are very subtle, leading to possible misclassification.
- Model performance heavily relies on X-ray quality (noise, resolution, angle). Poor-quality images can significantly reduce accuracy.The system uses only image data and ignores important clinical factors (age, symptoms, medical history), which are often crucial for accurate diagnosis.
- Advanced models like CNNs, EfficientNet, and Transformers require high processing power (GPU), making real-time or low-resource deployment challenging.
- Since the model is trained on a relatively limited dataset, it may learn patterns specific to the training data instead of general features, leading to reduced performance on new, unseen data.

1.5 SIGNIFICANCE OF THE STUDY

The significance of this study lies in developing an accurate and explainable deep learning framework for the automatic classification of knee osteoarthritis (KOA) using X-ray images. Knee osteoarthritis is one of the most common degenerative joint diseases, and early detection is essential to prevent severe joint damage and mobility loss. Traditional diagnosis methods mainly depend on radiologists' expertise, which can sometimes lead to subjective interpretation and delayed diagnosis. This research proposes a hybrid deep learning model that combines EfficientNetV2-S and ConvNeXt-Tiny architectures to improve classification performance by capturing both local texture features and global structural information from knee X-rays. The integration of attention mechanisms further enhances the model's ability to focus on clinically important regions, resulting in improved prediction accuracy of about 93%. Another important significance of the study is the use of Explainable Artificial Intelligence (XAI) through Grad-CAM visualization. This helps doctors understand which regions of the X-ray influenced the model's decision, thereby increasing trust and transparency in AI-assisted medical diagnosis. The study also contributes to the healthcare field by reducing diagnostic time, minimizing human error, and supporting clinicians in making faster and more reliable treatment decisions. Furthermore, the proposed system can serve as a cost-effective computer-aided diagnostic tool, especially in areas where experienced radiologists are limited.

SUMMARY

This study exploress on developing an automated deep learning system for classifying the severity of Knee Osteoarthritis (KOA) using X-ray images. It aims to overcome the limitations of the traditional Kellgren–Lawrence grading system, which can be subjective and inconsistent. Advanced deep learning models such as CNN, ResNet, DenseNet, and EfficientNet with Plug-in Modules are used to improve feature extraction and diagnostic accuracy. The project involves data collection, preprocessing, model training, and performance evaluation using metrics like accuracy, precision, recall, and F1-score. The proposed system helps provide faster and more reliable clinical decision-making. It also supports early diagnosis and effective treatment planning for patients. Overall, the study contributes to improving automated medical diagnosis and patient care.

CHAPTER 2

LITERATURE REVIEW

2.1 DEEP LEARNING BASED KOA STUDIES

[1] Tiulpin et al., 2018 proposed an automatic knee osteoarthritis diagnosis system using deep learning on plain knee radiographs. The model learned discriminative radiographic features directly from X-ray images without manual feature engineering. It achieved strong diagnostic performance in KL grading and OA detection tasks. The study demonstrated that convolutional neural networks can assist radiologists in large-scale OA screening. It also highlighted the importance of automated clinical decision support systems in orthopedic imaging.

[2] Tiulpin and Saarakkala, 2020 developed a deep CNN-based framework for automatic grading of individual KOA features from radiographs. Instead of predicting only overall severity, the model analyzed osteophytes, joint-space narrowing, and sclerosis independently. The system improved interpretability and clinical relevance of OA grading. Transfer learning and CNN feature extraction significantly enhanced classification accuracy. The work contributed toward fine-grained and explainable KOA assessment.

[3] Chen et al., 2020 introduced a deep learning severity grading model using knee X-ray images. The architecture extracted hierarchical image features to classify OA severity levels effectively. The proposed system reduced dependency on handcrafted image processing techniques. Experimental results showed improved accuracy compared with traditional machine learning approaches. The paper emphasized the role of automated grading systems in early OA diagnosis.

[8] Ghosh et al., 2021 focused on automated KOA detection using deep learning techniques applied to radiographic images. Different CNN architectures were evaluated to identify degenerative joint abnormalities. Image preprocessing and augmentation improved robustness against data imbalance. The framework demonstrated reliable OA classification performance for clinical applications. The authors suggested that AI-assisted systems can accelerate orthopedic diagnosis workflows.

[9] Tiulpin et al., 2020 summarized the progress and challenges of deep learning in knee osteoarthritis research. It discussed datasets, CNN architectures, transfer learning, and evaluation metrics used in OA analysis. The paper also highlighted limitations such as insufficient annotated datasets and poor model interpretability. Future directions included explainable AI and multimodal learning approaches. The review served as a comprehensive reference for KOA deep learning research.

[12] Ahmed and Imran, 2024 combined deep learning and explainable AI (XAI) for KOA analysis using X-ray images. The proposed framework not only classified osteoarthritis severity but also visualized important diagnostic regions. Explainability methods improved trustworthiness and clinical acceptance of the model. The system achieved high classification accuracy while maintaining transparency. The study emphasized interpretable AI for medical imaging applications.

[27] AI-Driven Osteoarthritis Diagnosis System, 2024 presented an AI-based diagnostic framework for clinical osteoarthritis detection. Deep learning models were trained on knee radiographs to classify disease severity automatically. The proposed approach improved diagnostic consistency and reduced manual assessment time. Clinical applicability and real-world deployment were major focuses of the study. The framework demonstrated the growing importance of AI-assisted healthcare systems.

2.2 EFFICIENT NET BASED STUDIES

[6] Tan and Le, 2021 – EfficientNetV2 introduced an optimized CNN architecture with faster training and improved parameter efficiency. The model combined neural architecture search with progressive learning techniques. It achieved state-of-the-art performance while reducing computational cost. EfficientNetV2 became highly suitable for medical imaging tasks due to its balanced accuracy and efficiency. Many KOA classification systems later adopted this architecture for improved feature extraction.

[22] Yeoh et al., 2023 proposed transfer learning-assisted 3D deep learning models for KOA detection using OAI data. EfficientNet-based architectures were utilized to extract volumetric knee joint features. Transfer learning improved model generalization even with limited labeled data. The framework achieved reliable classification performance in osteoarthritis detection tasks. The study demonstrated the usefulness of EfficientNet models for advanced medical imaging analysis.

2.3 DENSENET BASED STUDIES

[5] Thomas et al., 2020 implemented deep neural networks for automatic KOA severity classification. Dense connectivity concepts improved feature reuse and gradient propagation during training. The system successfully categorized multiple OA severity levels from X-ray images. Performance evaluation showed strong classification capability with reduced overfitting. The work demonstrated the effectiveness of DenseNet-inspired deep architectures in orthopedic imaging.

[10] Rajpurkar et al., 2017 CheXNet introduced a DenseNet-based architecture for radiologist-level chest X-ray diagnosis. Although focused on pneumonia detection, the model demonstrated the power of DenseNet in medical imaging applications. Dense feature propagation improved representation learning and classification accuracy. The success of CheXNet inspired similar DenseNet applications in KOA detection tasks. It became one of the foundational studies for medical deep learning research.

[11] Shen et al., 2017 reviewed deep learning applications in medical image analysis, including CNN and DenseNet-based methods. It explained how deep architectures automatically learn discriminative features from medical data. Dense connectivity and hierarchical learning improved image classification performance. The review also discussed challenges such as limited data and interpretability. It provided theoretical foundations for later KOA classification studies.

2.4 CONVNEXT BASED STUDIES

[13] Mehdi et al., 2025 proposed a shape-texture deep learning model for osteoarthritis scoring. ConvNeXt-based feature extraction improved texture representation and structural analysis of knee joints. The architecture captured both spatial and morphological abnormalities effectively. Experimental results showed improved OA severity prediction performance. The study highlighted ConvNeXt as a powerful alternative to traditional CNN architectures.

[17] Wang et al., 2025 introduced an attention-based deep learning model for osteoarthritis detection. ConvNeXt backbone networks combined with attention mechanisms enhanced focus on diseased joint regions. The framework improved feature discrimination and classification robustness. Results demonstrated superior performance compared with conventional CNN methods. The study emphasized attention-guided ConvNeXt architectures for precise OA diagnosis.

2.5 HYBRID DEEP LEARNING BASED STUDIES

[14] Srividhya et al., 2025 proposed a CNN-ELM framework for osteoarthritis classification. CNN layers extracted deep image features, while the Extreme Learning Machine performed rapid classification. The hybrid framework improved computational efficiency and prediction accuracy. Experimental results showed better performance than standalone CNN models. The study demonstrated the benefits of combining deep learning with fast classifiers.

[16] Srikijkasemwat et al., 2024 – KneeXNeT introduced an ensemble-based framework for radiographic knee evaluation. Multiple deep learning models were combined to improve robustness and generalization. The ensemble approach reduced classification errors caused by model bias. Performance evaluation showed superior OA grading accuracy compared with individual networks. The study highlighted the effectiveness of collaborative hybrid architectures.

[18] Mahum et al., 2025 proposed a hybrid deep learning framework for OA severity detection. Different CNN architectures were integrated to capture complementary image features. The hybrid system improved feature diversity and classification reliability. Experimental outcomes demonstrated enhanced severity prediction performance. The study confirmed that combining multiple architectures increases diagnostic robustness.

[20] Ren et al., 2025 – OA-MEN model fused ResNet and MobileNet architectures for osteoarthritis classification. ResNet captured deep semantic features while MobileNet reduced computational complexity. Feature fusion improved accuracy and efficiency simultaneously. The framework achieved strong performance on knee OA severity grading datasets. This work demonstrated the advantages of lightweight hybrid deep learning systems.

[21] Amjad et al., 2025 introduced an autoencoder-based framework for osteoarthritis classification. Autoencoders learned compressed latent representations of knee radiographs before classification. The hybrid design improved feature extraction and reduced noise in medical images. The framework achieved reliable OA severity prediction results. The paper emphasized unsupervised feature learning combined with deep classification techniques.

[23] Explainable Deep Learning for KOA Detection, 2024 integrated CNN models with explainable AI methods for osteoarthritis detection. Hybrid explainability techniques highlighted important regions responsible for predictions. The framework improved clinician trust and diagnostic transparency. Results showed accurate OA classification alongside interpretable visual explanations. The study promoted explainable hybrid AI systems for healthcare applications.

[24] Transfer Learning-Based OA Detection Framework, 2023 applied transfer learning with deep CNN architectures for OA detection. Pretrained models improved feature extraction from limited radiographic datasets. The hybrid transfer learning strategy reduced training time and improved accuracy. Experimental results validated the effectiveness of fine-tuned CNN models. The framework demonstrated efficient medical image classification using pretrained networks.

[25] Deep Learning-Based OA Severity Grading System, 2024 proposed a comprehensive deep learning framework for automated OA severity grading. Multiple CNN layers extracted radiographic features related to disease progression. The model improved grading consistency compared with manual assessment. Experimental results demonstrated reliable multi-class classification performance. The system supported accurate and efficient orthopedic diagnosis.

[26] CNN-Based OA Classification Model, 2025 developed a CNN architecture for automated osteoarthritis classification. Feature extraction and classification stages were optimized for radiographic image analysis. The framework achieved improved detection accuracy across multiple OA severity levels. Data augmentation and preprocessing enhanced model generalization. The work highlighted the effectiveness of deep CNN models in orthopedic imaging.

[28] Yoon et al., 2023 developed a deep learning software tool for automatic feature extraction and grading of radiographic KOA. The system automatically analyzed knee X-rays and predicted disease severity. Fine-grained feature learning improved diagnostic precision and consistency. The software demonstrated strong clinical applicability in orthopedic assessment. The study contributed toward automated and user-friendly AI diagnostic systems.

[29] Pi et al., 2023 proposed ensemble deep learning networks for automated KOA grading. Multiple CNN architectures were combined to improve classification reliability. Ensemble learning reduced variance and enhanced prediction stability. Results demonstrated higher grading accuracy than individual deep models. The framework showed strong potential for real-world clinical deployment.

[30] Abd El-Ghany et al., 2023 introduced a fully automatic fine-tuned deep learning model for KOA detection and progression analysis. Transfer learning and fine-tuning improved feature extraction from radiographic images. The framework effectively classified OA severity and monitored disease progression. Experimental results showed strong predictive performance with minimal manual intervention. The study emphasized automation and efficiency in AI-based orthopedic diagnosis.

2.7 GAP IN EXISTING RESEARCH

Despite significant advancements in deep learning techniques for Knee Osteoarthritis (KOA) classification, several important research gaps still remain. Earlier studies mainly focused on traditional convolutional neural network (CNN) architectures such as AlexNet, VGG16, ResNet, DenseNet, and EfficientNet for automatic OA grading using knee X-ray images.

Although these methods achieved moderate to high accuracy, many models struggled to provide consistent performance in multi-class classification tasks, especially when distinguishing intermediate severity levels such as Osteopenia and Osteoporosis. Several studies reported classification accuracies between 66% and 85%, showing limitations in feature representation and generalization. Existing KOA classification systems using single backbone architectures struggle to capture both local and global image features effectively, highlighting the need for a robust hybrid framework to improve feature extraction, accuracy, and computational efficiency.Many previous studies rely on limited and less diverse datasets, which may not reflect real-world clinical variability and can reduce model generalization. Dataset imbalance also affects prediction performance by creating bias toward majority classes, and although techniques like weighted sampling, data augmentation, and transfer learning are used, reliable performance across diverse patient populations remains a challenge.

Interpretability remains a major research gap in current Knee Osteoarthritis (KOA) classification systems because most deep learning models operate as black-box frameworks with unclear decision-making processes.Although many CNN and hybrid architectures achieve high classification accuracy, they often fail to provide transparent explanations for their predictions.This lack of interpretability reduces clinical trust, making it difficult for radiologists and healthcare professionals to rely completely on AI-assisted diagnosis.Some recent studies have attempted to address this issue using explainable AI (XAI) techniques such as Grad-CAM, attention maps, and saliency visualization methods. These techniques help identify important regions in knee X-ray images that influence model predictions and improve understanding of disease-related features.

Hence, the proposed hybrid deep learning framework combines efficient feature extraction, attention mechanisms, transfer learning, explainable AI, and balanced learning strategies for accurate and interpretable multi-class KOA classification. By integrating EfficientNetV2-S, ConvNeXt-Tiny, attention-based feature fusion, and Grad-CAM visualization, the model improves both classification performance and clinical interpretability in knee osteoarthritis diagnosis.

SUMMARY

This chapter reviewed various deep learning techniques used for Knee Osteoarthritis (KOA) classification using knee X-ray images. Earlier studies mainly used CNN architectures such as ResNet, DenseNet, EfficientNet, and ConvNeXt for automatic feature extraction and OA severity grading. Researchers also applied transfer learning, ensemble learning, attention mechanisms, hybrid models, and explainable AI techniques to improve classification accuracy and interpretability. However, existing systems still face challenges such as poor multi-class classification performance, dataset imbalance, limited generalization, and lack of transparency in decision-making. To overcome these limitations, the proposed work introduces a hybrid framework combining EfficientNetV2-S, ConvNeXt-Tiny, attention-based feature fusion, transfer learning, balanced learning strategies, and Grad-CAM visualization for accurate and interpretable KOA diagnosis.

CHAPTER 3

SYSTEM SPECIFICATION

3.1 HARDWARE SPECIFICATION

- PROCESSOR : Intel Core i5 / i7 or AMD Ryzen 5 / Ryzen 7
- RAM : Minimum 8 GB (16 GB Recommended)
- STORAGE : 256 GB SSD or higher
- GPU : NVIDIA GPU with CUDA support

3.2 SOFTWARE SPECIFICATION

- OPERATING SYSTEM : Windows 10/11 or Ubuntu Linux
- PROGRAMMING LANGUAGE : Python 3.x
- DEEP LEARNING FRAMEWORK : TensorFlow, Keras, PyTorch
- DEVELOPMENT TOOLS : Google Colab, VS Code
- DATA COLLECTION TOOLS : Kaggle

3.3 TOOLS DESCRIPTION

3.3.1 PYTHON

Python is a high-level, interpreted programming language widely used in the field of artificial intelligence, machine learning, and deep learning. It is popular because of its simple syntax, readability, and extensive library support, making it suitable for both beginners and advanced developers. In this project, Python is used to develop and train deep learning models for the classification of knee osteoarthritis using X-ray images. Python supports powerful frameworks such as TensorFlow, Keras, and PyTorch, which help in building convolutional neural networks and transfer learning models like EfficientNet, DenseNet, and ResNet50V2. The language also provides libraries such as NumPy and Pandas for numerical computation and dataset handling. OpenCV and PIL are used for image preprocessing and enhancement tasks. Matplotlib and Seaborn assist in data visualization, enabling clear representation of training accuracy, loss graphs, and confusion matrices. Python allows seamless integration of medical image processing and deep learning workflows within a single environment. Its large community support and open-source nature make it highly reliable for research and academic projects.

3.3.2 Google Colab

Google Colab is a cloud-based Jupyter Notebook environment used for executing Python programs and deep learning projects.It provides free GPU and TPU support, which improves the speed and efficiency of model training.In this project, Colab is used for knee osteoarthritis classification using Knee X-ray datasets.It supports popular deep learning libraries such as TensorFlow, Keras, and PyTorch for implementing neural network models.Google Colab also enables easy dataset storage and sharing through integration with Google Drive and supports real-time collaboration.The platform helps efficiently train and evaluate models like EfficientNetB3, DenseNet121, and ResNet50V2 with visualization and automatic saving features.

3.3.3 Kaggle

Kaggle is an online platform widely used for data science, machine learning, and deep learning research.In this project, Kaggle provides Knee X-ray datasets for training and testing knee osteoarthritis classification models.It offers cloud-based notebook environments with GPU support, enabling faster execution of deep learning algorithms.Kaggle supports Python and popular libraries such as TensorFlow, Keras, PyTorch, NumPy, and Pandas.The platform helps in dataset preprocessing, model experimentation, performance evaluation, and transfer learning implementation.Kaggle notebooks allow organized documentation of code, outputs, graphs, and explanations while supporting collaboration and knowledge sharing.Overall, Kaggle serves as an efficient platform for dataset acquisition, experimentation, and implementation of models like EfficientNetB3, DenseNet121, and ResNet50V2.

SUMMARY

This chapter explains the hardware and software requirements for developing the knee osteoarthritis classification system using deep learning.It includes hardware components like processors, GPU support, RAM, and SSD storage, along with software tools such as Python, TensorFlow, Keras, Google Colab, VS Code, and Kaggle.

CHAPTER -4

DATASET AND METHODOLY

4.1 DATA COLLECTION

The Knee Osteoarthritis Classification [224×224] dataset was collected for multi-class classification of knee X-ray images into Normal, Osteopenia, and Osteoporosis categories. The dataset was created and published by Fuyad Hasan Bhoyan and Prashanta Sarker on Kaggle. It contains knee radiographic images gathered from multiple reliable medical imaging sources, including the Osteoarthritis Initiative (OAI) and Mendeley Data repositories. The images were resized to 224×224 pixels and enhanced using augmentation techniques such as flipping and rotation to balance class distribution and improve model generalization.

4.2 DATASET DESCRIPTION

The dataset used in this study is the Knee Osteoarthritis Classification dataset developed for automated image classification using knee X-ray radiographs. The dataset contains a total of 5400 knee X-ray images organized into training, validation, and testing subsets to support effective deep learning model development and evaluation. Among these, 3780 images are allocated for training, 1080 images for validation, and 540 images for final testing and performance analysis. The dataset contains three balanced classes Normal, Osteoporosis, and Osteopenia with approximately 1260 images in each category . The balanced distribution of images supports stable learning and reliable multi-class classification performance. The dataset is stored in image file format using medical knee radiographs suitable for preprocessing, augmentation, feature extraction, and deep learning analysis. The validation set supports model tuning and overfitting control, while the test set evaluates final model performance and accuracy. The dataset structure supports advanced deep learning architectures such as EfficientNetB3, DenseNet121, ResNet50V2, ConvNeXt, and hybrid deep learning frameworks for KOA severity classification. Table 4.1 illustrates the important dataset features including dataset name, dataset type, total images, subset organization, class distribution, data balance, validation purpose, and testing purpose used in this research is

https://www.kaggle.com/datasets/fuyadhasanbhoyan/knee-osteoarthritis-classification-224224

TABLE 4.1 Knee Osteoarthritis Classification Dataset Features and Structure

S. No	Feature	Description	Values
1	Dataset Name	Name of the dataset used for the study	Knee Osteoarthritis Classification
2	Dataset Type	Type of machine learning problem	Image Classification
3	Total Images	Total number of images in the dataset	5400
4	Data Structure	Organization of dataset into subsets	Train, Validation, Test
5	Training Set Size	Number of images used for training	3780
6	Validation Set Size	Number of images used for validation	1080
7	Test Set Size	Number of images used for testing	540
8	Number of Classes	Total categories in classification task	3
9	Class 1	First category of images	Normal (1260 images)
10	Class 2	Second category of images	Osteoporosis (1260 images)
11	Class 3	Third category of images	Osteopenia (1260 images)
12	Data Balance	Distribution of images across classes	Balanced
13	Data Format	Type of data used	Image files (X-ray images)
14	Validation Purpose	Role of validation set	Hyper-parameter tuning
15	Test Purpose	Role of test set	Final model evaluation

4.3 RESEARCH DESIGN

4.3.1 Research Methodology and Model Development

This study follows an applied quantitative research approach using deep learning techniques for automated Knee Osteoarthritis (KOA) classification from knee X-ray images. A data-driven methodology is adopted where medical imaging data is used to extract meaningful features and predict osteoarthritis severity levels automatically. The research develops CNN-based architectures such as EfficientNet, DenseNet, and ResNet integrated with plug-in modules and ensemble learning techniques to improve classification accuracy and robustness. Transfer learning using pretrained ImageNet weights is employed for efficient feature extraction, while ensemble methods combine predictions from multiple models to enhance overall diagnostic performance and reliability.

4.3.2 Data Collection and Preprocessing

This study collects labeled knee X-ray images from publicly available medical imaging datasets such as Kaggle for automated Knee Osteoarthritis (KOA) classification. The dataset includes images representing Normal, Osteoporosis, and Osteopenia conditions used for training and testing deep learning models. Data preprocessing techniques such as image resizing to 224×224 pixels, normalization, and augmentation are applied to improve dataset quality and consistency. Augmentation methods including rotation, zooming, and flipping help increase data diversity, reduce overfitting, and enhance the learning capability and generalization performance of deep learning models. Preprocessing also removes inconsistencies and improves feature visibility in knee X-ray images for accurate classification.

4.3.3 Model Training, Evaluation, and Statistical Analysis

This study trains deep learning models using transfer learning techniques and evaluates their performance using metrics such as accuracy, precision, recall, and F1-score. The trained models demonstrate strong classification performance across multiple osteoarthritis severity levels, while validation and testing procedures ensure robustness and generalization capability. Statistical analysis is performed using confusion matrices and performance evaluation metrics to validate model effectiveness and reliability. These measures help compare the strengths and limitations of different deep learning architectures and ensure consistent, accurate, and reliable prediction results**.**

4.3.4 Result Interpretation and Future Enhancement

This study analyzes model outputs to evaluate classification accuracy across different osteoarthritis severity levels and demonstrates the capability of deep learning models to identify subtle variations in knee X-ray images. The experimental results indicate that hybrid deep learning architectures can effectively improve feature extraction, classification robustness, and prediction consistency. Performance metrics such as accuracy, precision, recall, and F1-score confirm the effectiveness of the proposed models in automated KOA classification tasks. Future enhancements may include the use of larger and more diverse datasets, integration of multi-modal clinical and imaging data, and advanced explainable AI techniques such as Grad-CAM and SHAP to improve prediction accuracy, interpretability, and clinical trust in automated KOA diagnosis systems.

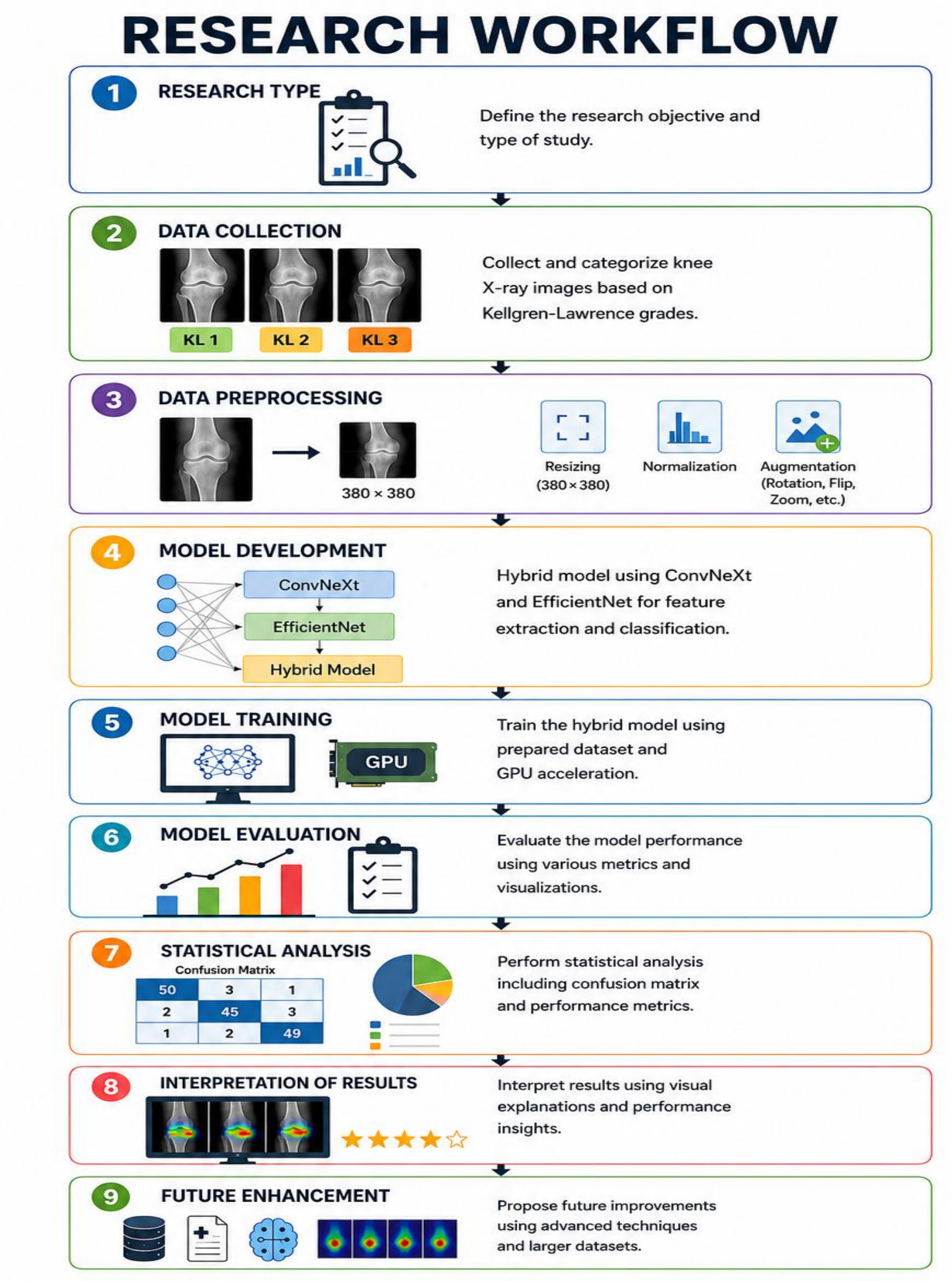


Fig 4.1 RESEARCH DESIGN

Fig 4.1 illustrates the complete workflow for Knee Osteoarthritis classification using deep learning techniques.

4.4 METHODOLOGY

The workflow of the proposed model flow is as follows:

Fig 4.2 PROPOSED MODEL WORKFLOW

Figure 4.2 illustrates the workflow for Knee Osteoarthritis (KOA) severity prediction using deep learning models.

4.4.1 Knee Osteoarthritis Detection and Classification Workflow

The proposed system uses Knee Osteoarthritis (KOA) X-ray image datasets with advanced deep learning models such as DenseNet121, EfficientNet, ResNet50V2, and hybrid architectures to improve automated severity classification accuracy and robust feature extraction. During preprocessing, the Knee X-ray images are enhanced by reducing noise, correcting contrast variations, blur, and imaging inconsistencies, while corrupted and duplicate images are removed to improve dataset quality. The images are resized to standardized dimensions, normalized for stable training, and augmented to increase dataset diversity and reduce overfitting. The processed dataset is then divided into training and testing sets for effective learning and evaluation. Finally, deep learning models automatically extract important radiographic features such as joint space narrowing, bone texture, osteophytes, cartilage degradation, and structural abnormalities to accurately classify and predict Knee Osteoarthritis severity.

4.4.2 Deep Learning Models for Knee Osteoarthritis Classification

DenseNet, ResNet, and EfficientNet are used with transfer learning and pre-trained ImageNet weights for accurate Knee Osteoarthritis severity classification from X-ray images. DenseNet121 improves feature reuse and gradient flow through dense connections, enabling effective learning of detailed radiographic patterns while reducing vanishing gradient issues. ResNet50V2 utilizes residual and skip connections to support deep feature extraction, efficient training, and accurate identification of subtle structural changes in knee joints. EfficientNetB3 employs compound scaling and an optimized architecture to achieve high classification accuracy with lower computational complexity, enabling efficient extraction of fine-grained radiographic features from knee X-ray images. These deep learning models automatically learn important medical image features without requiring manual feature engineering, improving the efficiency of disease diagnosis. The combination of transfer learning and advanced neural network architectures enhances model generalization and robustness when classifying unseen Knee X-ray images. Together, these models provide reliable, scalable, and high-performance solutions for automated Knee Osteoarthritis detection and severity assessment in medical imaging applications.

4.4.3 Hybrid Learning and KOA Severity Prediction

The hybrid deep learning model combines EfficientNet EfficientNetV2-S and ConvNeXt ConvNeXt-Tiny with preprocessing, data augmentation, normalization, and weighted sampling techniques to improve the accuracy, robustness, and reliability of Knee Osteoarthritis classification from X-ray images. The trained models automatically classify Knee X-ray images into different KOA severity levels such as Normal, Osteopenia, and Osteoporosis by learning important radiographic patterns and structural abnormalities in knee joints. This automated prediction system supports early diagnosis, effective treatment planning, continuous disease monitoring, and efficient large-scale screening while reducing dependence on manual radiographic interpretation and improving clinical decision-making accuracy.

4.4.4 Model Evaluation

The performance of the models is evaluated using standard classification metrics

Accuracy- Measures overall prediction correctness.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision- Measures positive prediction correctness.

$$Precision = \frac{TP}{TP + FP}$$

Recall - Measures the ability to identify true positive cases.

$$Recall = \frac{TP}{TP + FN}$$

F1-Score - Balances precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

4.4.5 Performance Evaluation and Clinical Decision Support

Confusion matrix analysis is used to evaluate the Knee Osteoarthritis prediction system by identifying correct and incorrect classifications across different severity classes, helping measure model accuracy, interpretability, and classification weaknesses. Various visualization techniques such as accuracy graphs, loss curves, confusion matrices, and classification reports are utilized to analyze and compare the performance of deep learning models in the KOA classification system. The final stage of the system functions as an intelligent clinical decision support system by integrating hybrid feature fusion, ensemble learning, and automated KOA severity classification to improve diagnostic accuracy and reliability. This approach reduces manual radiographic interpretation effort, supports early disease detection, enhances treatment planning, and assists healthcare professionals in making faster and more effective clinical decisions.

4.5 Hybrid Deep Learning Model Architecture

The hybrid knee osteoarthritis classification system is developed using the PyTorch framework by combining EfficientNetV2-S and ConvNeXt Tiny models for accurate X-ray image classification. The dataset is preprocessed using resizing, normalization, and augmentation techniques, while WeightedRandomSampler handles class imbalance during training. The fused features from EfficientNetV2-S and ConvNeXt Tiny are processed through an attention-based classifier and optimized using AdamW, mixed precision training, and early stopping, while performance is evaluated using accuracy, confusion matrix, ROC-AUC, precision, recall, and F1-score metrics.

4.6 DENSENET121 MODEL

DenseNet121 is a deep learning model that enhances Knee Osteoarthritis classification from X-ray images through dense connectivity, which improves feature reuse, information flow, and gradient propagation for accurate OA grading. In the proposed workflow, essential libraries such as TensorFlow, Keras, NumPy, Matplotlib, and scikit-learn are used for image processing, evaluation, and visualization. A pretrained DenseNet121 model is applied through transfer learning to extract meaningful features from Knee X-ray images, while ImageDataGenerator and DenseNet preprocessing techniques resize images to 224×224 pixels and normalize pixel values. Dense blocks, convolution layers, and transition layers improve learning efficiency and reduce model complexity, followed by global average pooling and fully connected layers for final classification. The trained model predicts osteoarthritis severity classes, and performance is evaluated using accuracy, precision, recall, F1-score, classification reports, confusion matrices, and normalized confusion matrices to analyze classification effectiveness across different categories.

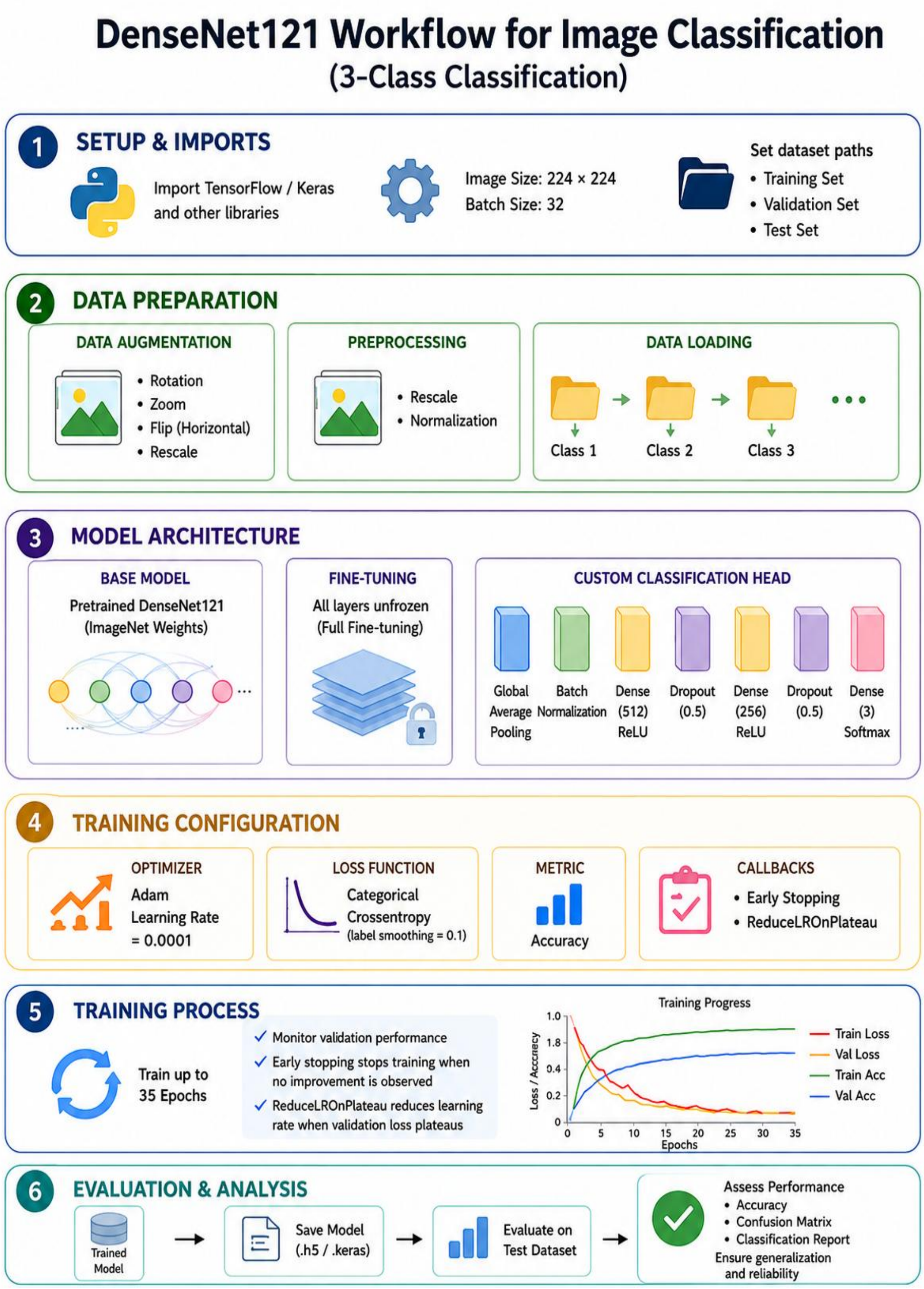


Fig 4.3 PROCESS OF DENSETNET 121 MODEL

Figure 4.3 illustrates the complete workflow of the DenseNet121 model, including data preprocessing, augmentation, feature extraction, training, and evaluation processes.

4.7 EFFICIENTNET B3 MODEL

EfficientNet is a deep learning CNN model that uses compound scaling of depth, width, and image resolution to achieve high classification accuracy with reduced computational cost and fewer parameters. In the proposed system, Knee X-ray images are first preprocessed through resizing, normalization, and augmentation using ImageDataGenerator to improve robustness and generalization. A pretrained EfficientNetB3 model with ImageNet weights is then applied for efficient feature extraction from medical images. Additional layers including GlobalAveragePooling2D, Batch Normalization, Dense, Dropout, and Softmax are integrated for three-class classification. The model is trained using the Adam optimizer with categorical cross-entropy loss, while EarlyStopping and ReduceLROnPlateau callbacks help minimize overfitting and optimize performance during training for up to 40 epochs.

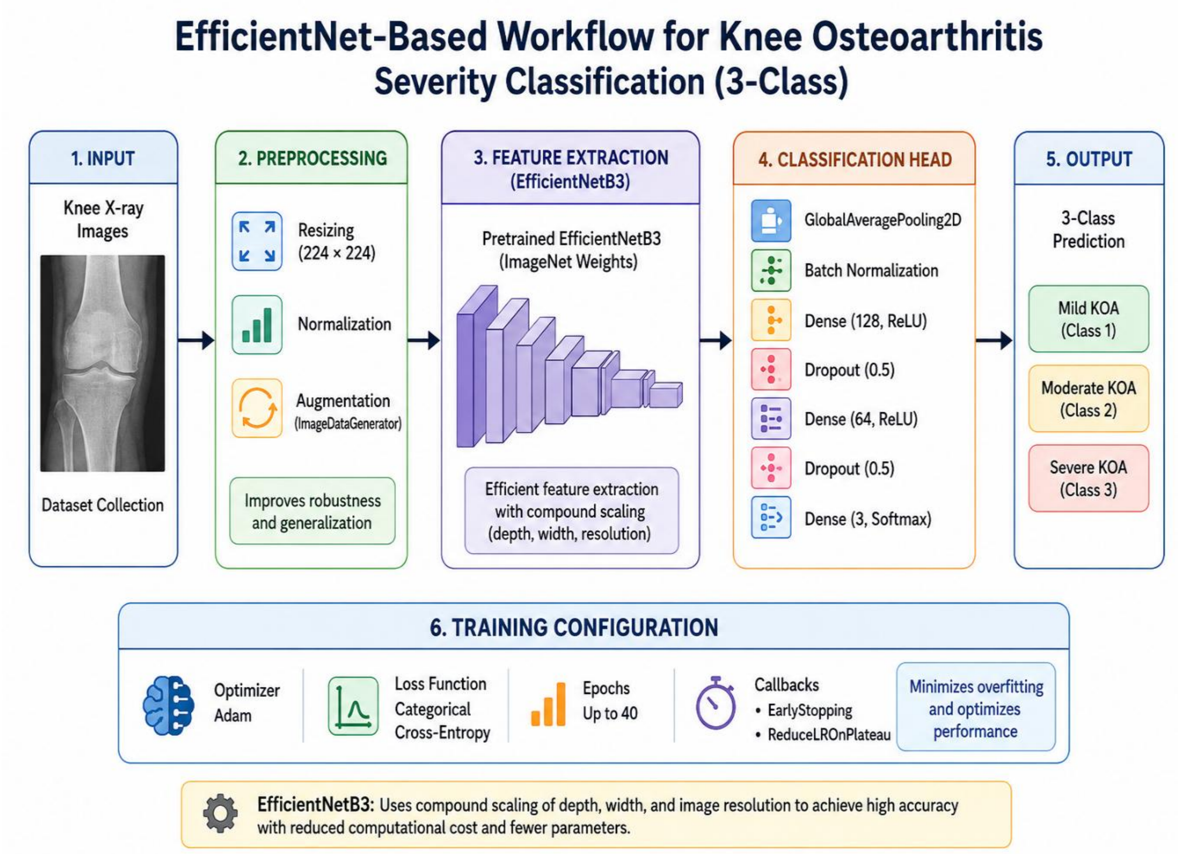


Fig 4.4 PROCESS OF EFFICIENT NET B3 MODEL

Fig4.4 illustrates the EfficientNetB3 workflow includes image preprocessing, feature extraction, optimized model training, and final classification using a Softmax layer.

4.8 RESTNET50 V2 MODEL

ResNet improves image classification performance by using convolution layers and residual skip connections that reduce the vanishing gradient problem, allowing very deep models such as ResNet50, ResNet101, and ResNet152 to train effectively with higher accuracy and stable learning. In the proposed workflow, Knee X-ray images are first prepared and augmented using ImageDataGenerator to improve model generalization and minimize overfitting. A pretrained ResNet50V2 model is then combined with custom classification layers for three-class image classification. The network is trained using the Adam optimizer for 40 epochs with EarlyStopping and ReduceLROnPlateau callbacks to enhance training stability and optimize performance. Finally, the model is evaluated using classification reports, confusion matrices, accuracy-loss curves, and ROC-AUC analysis to measure overall prediction accuracy and reliability.

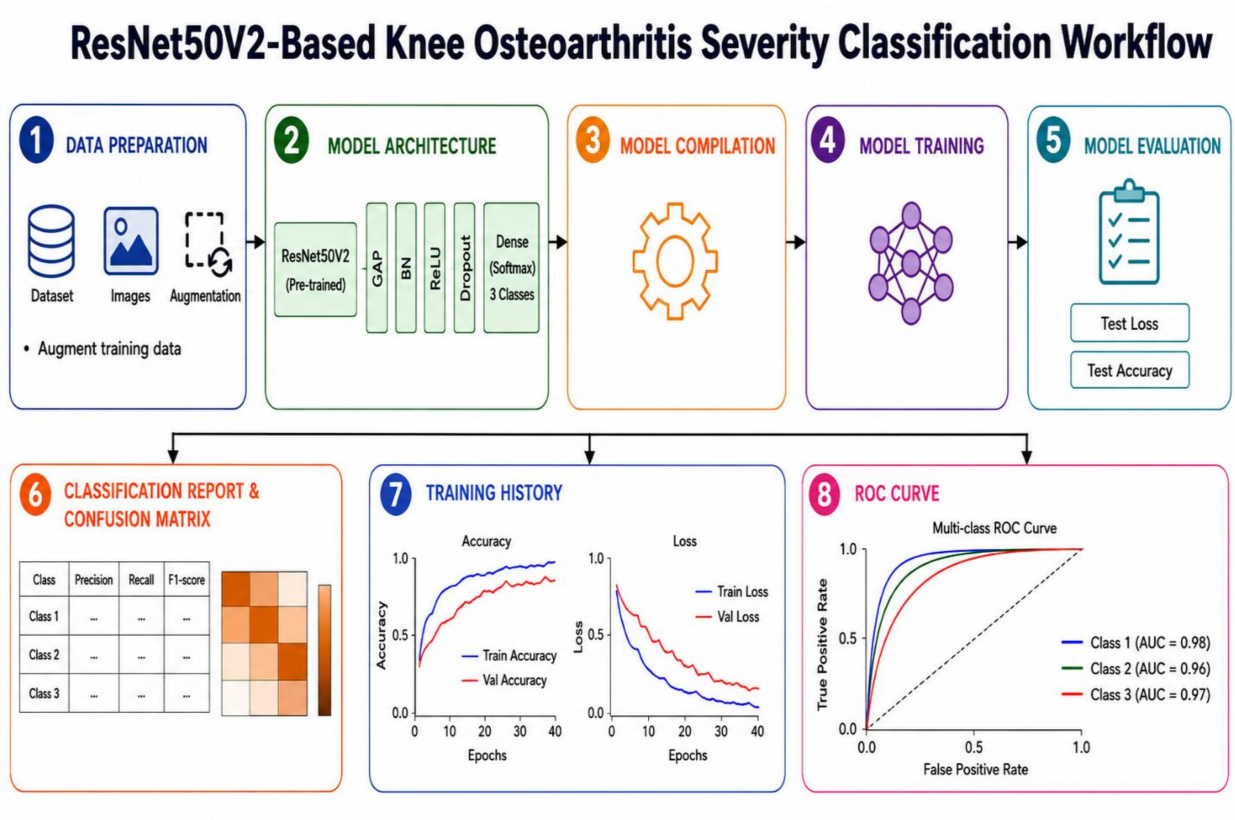


Fig 4.5 PROCESS OF REST NET 50 V2 MODEL

Fig4.5 illustrates a complete deep learning workflow beginning with data preparation and augmentation using ImageDataGenerator, followed by a pre-trained ResNet50V2 model

4.9 HYBRID MODEL

The hybrid deep learning model is designed to improve Knee Osteoarthritis (KOA) severity classification by combining the strengths of multiple deep learning architectures. Unlike traditional single-model approaches, hybrid models can capture diverse and complementary image features, resulting in better classification accuracy and robustness. In the proposed system, Knee X-ray images are initially preprocessed using normalization, resizing, and augmentation techniques to improve image quality and increase dataset diversity. Additionally, the WeightedRandomSampler technique is applied to handle class imbalance problems, ensuring that all severity classes are equally represented during training and preventing biased predictions.

The architecture of the proposed system integrates pretrained EfficientNetV2-S and ConvNeXt-Tiny models as backbone feature extractors. EfficientNetV2-S efficiently captures fine-grained radiographic details with optimized computational performance, while ConvNeXt-Tiny enhances deep feature representation through modern convolutional network design. A custom attention mechanism is incorporated to focus on important regions of Knee X-ray images, such as joint space narrowing, osteophytes, cartilage degradation, and bone structural abnormalities. The extracted features from both models are fused together to generate a more informative and discriminative feature representation before classification. Finally, fully connected layers and a Softmax activation function are used to classify the Knee Osteoarthritis severity into different classes.

To improve training stability and reduce overfitting, the model uses CrossEntropyLoss with label smoothing, which prevents overconfident predictions and improves generalization performance. The AdamW optimizer is applied to achieve efficient weight optimization, while the ReduceLROnPlateau scheduler dynamically adjusts the learning rate based on validation performance. Early stopping is also implemented to stop training automatically when no significant improvement is observed, reducing unnecessary computation and preventing model overfitting. These optimization strategies help the model achieve better convergence and reliable performance during training.

Furthermore, Grad-CAM visualizations and attention heatmaps provide interpretability by highlighting the important regions of Knee X-ray images that influence the model’s predictions. These visualization techniques improve the transparency and reliability of the system, making the proposed hybrid deep learning model highly effective for accurate and interpretable Knee Osteoarthritis severity classification.

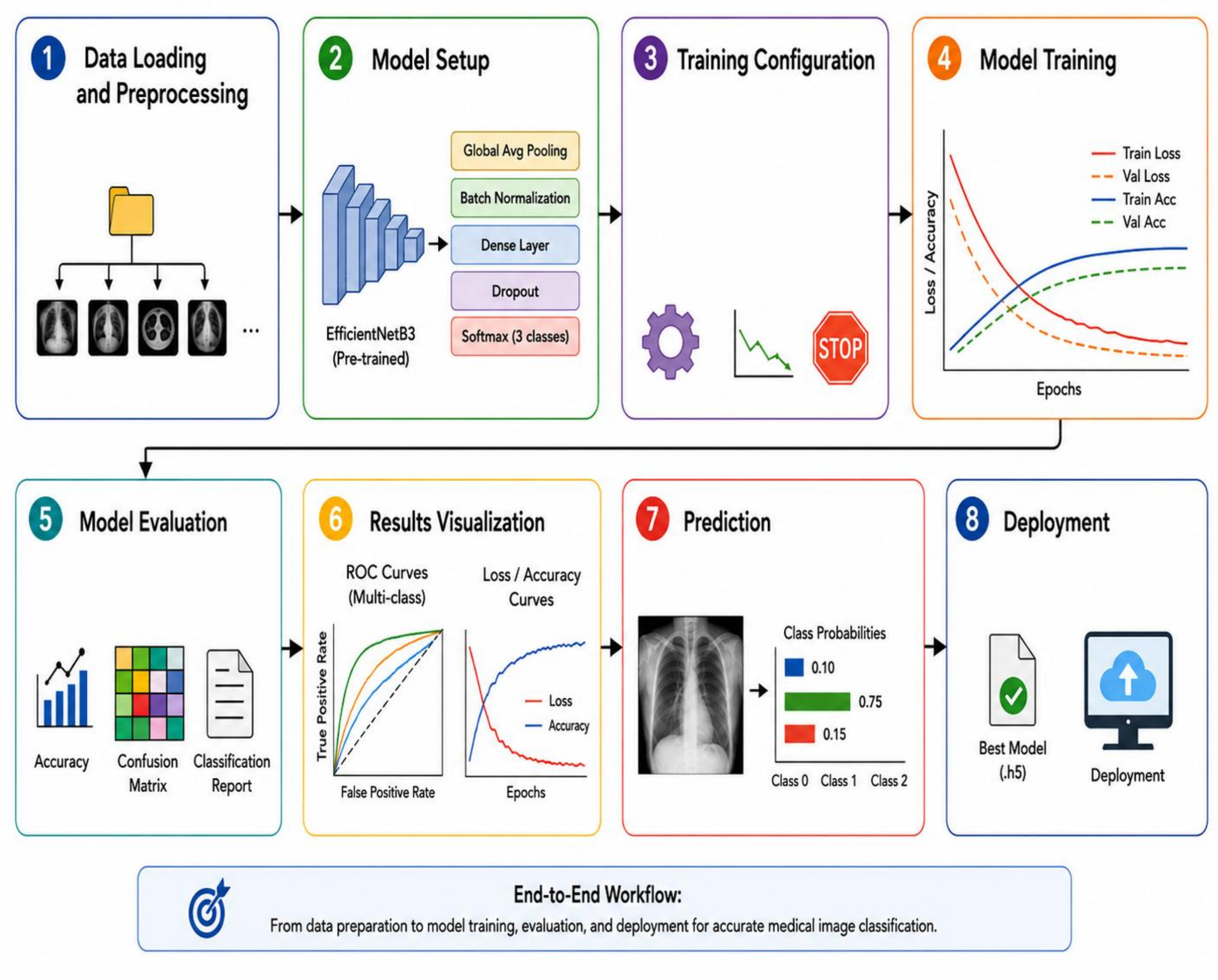


Fig 4.6 PROCESS OF HYBRID MODEL

Fig4.6 illustrates the complete EfficientNetB3 workflow for medical image classification.

SUMMARY

This chapter presents the complete dataset description and methodological framework for Knee Osteoarthritis (KOA) classification using deep learning techniques. The dataset consists of knee X-ray images collected from Kaggle Kaggle and labeled according to the Kellgren–Lawrence grading system Kellgren–Lawrence grading system, covering different severity levels from Normal to Severe.The images are preprocessed through resizing, normalization, cleaning, and augmentation techniques such as rotation, flipping, zooming, and shifting to improve data quality and reduce overfitting.

CHAPTER -5

RESULT AND DESCUSSION

Knee Osteoarthritis (KOA) is a major joint disorder that affects mobility and daily life. The proposed system uses deep learning and multiple CNN architectures to automatically detect and classify KOA from knee X-ray images. Image preprocessing techniques such as resizing, normalization, and data augmentation were applied to enhance dataset quality and reduce overfitting. Transfer learning models including ResNet50V2, DenseNet121, EfficientNetB3, and a Hybrid model were evaluated using accuracy, precision, recall, and F1-score for reliable KOA severity classification.

5.1 PERFORMNCE METRICS

$$\boldsymbol{Accuracy{=}TP{+}TN{+}FP{+}FNTP{+}TN} \quad \text{---}\mathbf{eq(1)}$$

Equation-1 explains Accuracy measures how correctly the model predicts both positive and negative cases out of all predictions made.It is calculated as the ratio of correctly classified instances (TP + TN) to the total number of cases (TP + TN + FP + FN).

$$\boldsymbol{Precision = (TP) / (TP + FP)} \quad \mathbf{--eq(2)}$$

Equation(2) explains Precision measures how many of the samples predicted as positive are actually correct. It is calculated as the ratio of true positives (TP) to the total predicted positives (TP + FP), where FP represents false positives.

$$\boldsymbol{Recall = (TP) / (TP + FN)} \quad \mathbf{--eq(3)}$$

Recall measures how well the model correctly identifies actual positive cases out of all real positives.It is the ratio of true positives (TP) to the sum of true positives and false negatives (TP + FN).

$$\boldsymbol{F1 - Score = 2 * (Precision * Recall) / (Precision + Recall)} \quad \mathbf{–eq(4)}$$

F1-score is a performance metric that combines precision and recall into a single value to measure a model's accuracy, especially in imbalanced datasets

$$\boldsymbol{TPR{=}TP{+}FNT} \quad \mathbf{--eq(5)}$$

True Positive Rate (TPR), also called Recall or Sensitivity, is calculated as **TP / (TP + FN)**. It measures the proportion of actual positive cases correctly identified by the model.

ROC Curve = (FPR, TPR) **--equ(6)**

ROC Curve evaluates the performance of a classification model by comparing the True Positive Rate (TPR) and False Positive Rate (FPR) at different threshold levels.

5.2 EXPERIMENTAL SETUP FOR THE MODELS

The experimental setup for all models (EfficientNetB3, DenseNet121, ResNet50V2, and the hybrid architecture) involves training on knee osteoarthritis X-ray images categorized using the Kellgren–Lawrence grading system, with a standard preprocessing pipeline including resizing of images, normalization, and data augmentation techniques such as rotation, flipping, zooming, and shifting to improve generalization. The dataset is typically split into training, validation, and testing sets to ensure unbiased evaluation. All models are initialized with ImageNet pre-trained weights and fine-tuned by replacing the top layers with custom fully connected layers for multi-class classification. The models are trained using the Adam-W with categorical cross-entropy loss, a fixed learning rate schedule, and batch sizes optimized for GPU memory efficiency. The model is trained with callbacks like EarlyStopping and ReduceLROnPlateau to reduce overfitting and improve convergence, while performance is evaluated using metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC analysis.

5.2.1 DenseNet121 Performance Metrics

The DenseNet121 model used transfer learning with ImageNet weights for knee osteoarthritis classification, along with image preprocessing and augmentation to improve generalization. The model achieved reliable performance using the AdamW, categorical cross-entropy loss, and evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix. DenseNet introduces dense connections where each layer receives feature maps from all previous layers:

$$\boldsymbol{x_l = H_l([x_0, x_1, x_2, \dots, x_{l-1}]) - -eq(6)}$$

where $H_l(\cdot)$represents a composite function of Batch Normalization, ReLU, and

Convolution, means each DenseNet layer takes as input the concatenation of feature maps from all previous layers and applies a transformation $H_l$to produce its output.

Table 5.1 DenseNet121 Performance Metrics

Class	Precision (%)	Recall (%)	F1-Score (%)
Normal	93%	83%	88%
Osteopenia	87%	97%	92%
Osteoporosis	83%	83%	83%

5.2.2 EfficientNetB3 Performance Metrics

The deep learning models were evaluated using accuracy, precision, recall, F1-score, and confusion matrix metrics to measure classification performance. The results confirmed that EfficientNetB3, DenseNet121, and ResNet effectively and reliably classified Normal, Osteopenia, and Osteoporosis conditions from Knee Osteoarthritis X-ray images.

EfficientNetB3 uses the compound scaling formula

$$\textbf{Depth} = \boldsymbol{\alpha^{\phi}}, \textbf{ Width} = \boldsymbol{\beta^{\phi}}, \textbf{ Resolution} = \boldsymbol{\gamma^{\phi}} \textbf{where } \boldsymbol{\alpha \times \beta^{2} \times \gamma^{2} \approx 2.} \textbf{ –eq(7)}$$

This formula balances network depth, width, and image resolution together to improve model accuracy and efficiency while reducing computational cost.

Table 5.2 EfficientNetB3 Performance Metrics

Class	Precision (%)	Recall (%)	F1-Score (%)
Normal	94%	93%	93%
Osteopenia	92%	95%	93%
Osteoporosis	91%	89%	90%

5.2.3 ResNet50 V2 Performance Metrics

The performance of the ResNet50V2 model was evaluated using accuracy, precision, recall, and F1-score metrics for knee osteoarthritis classification. These metrics measured the model’s prediction accuracy, disease detection capability, and balanced classification performance, demonstrating its reliability and robustness in medical image analysis. ResNet50V2 uses the residual learning formula

$$\boldsymbol{y} = \boldsymbol{F}(\boldsymbol{x}, \boldsymbol{W}) + \boldsymbol{x} \quad \text{—eq(8)}$$

Here, $F(x, W)$represents the learned residual mapping and $x$is the input added through a skip connection, which helps reduce vanishing gradient problems and improves deep network training performance.

Table 5.3 ResNet50 V2 Performance Metrics

Class	Precision (%)	Recall (%)	F1-Score (%)
Normal	88%	86%	87%
Osteopenia	86%	90%	88%
Osteoporosis	86%	83%	84%

5.2.4 HYBRID MODEL Performance Metrics

Performance metrics evaluate the effectiveness of the deep learning model using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix analysis to measure classification accuracy, disease detection capability, and overall balanced performance.

$$\boldsymbol{H}(\boldsymbol{x}) = \boldsymbol{\alpha D}(\boldsymbol{x}) + \boldsymbol{\beta R}(\boldsymbol{x}) \qquad \text{--eq(9)}$$

Here, $D(x)$features extracted from DenseNet121 and ResNet50V2 are combined using weighted fusion, where α and β control the contribution of each model.

Table 5.4 Hybrid Performance Metrics

Class	Precision (%)	Recall (%)	F1-Score (%)
Normal	88%	91%	90%
Osteopenia	88%	93%	91%
Osteoporosis	87%	79%	83%

5.3 DISCUSSION OF RESULT

5.3.1 DenseNet121 Model

The DenseNet model output shows the training and validation accuracy/loss performance over multiple epochs during knee osteoarthritis classification**.**

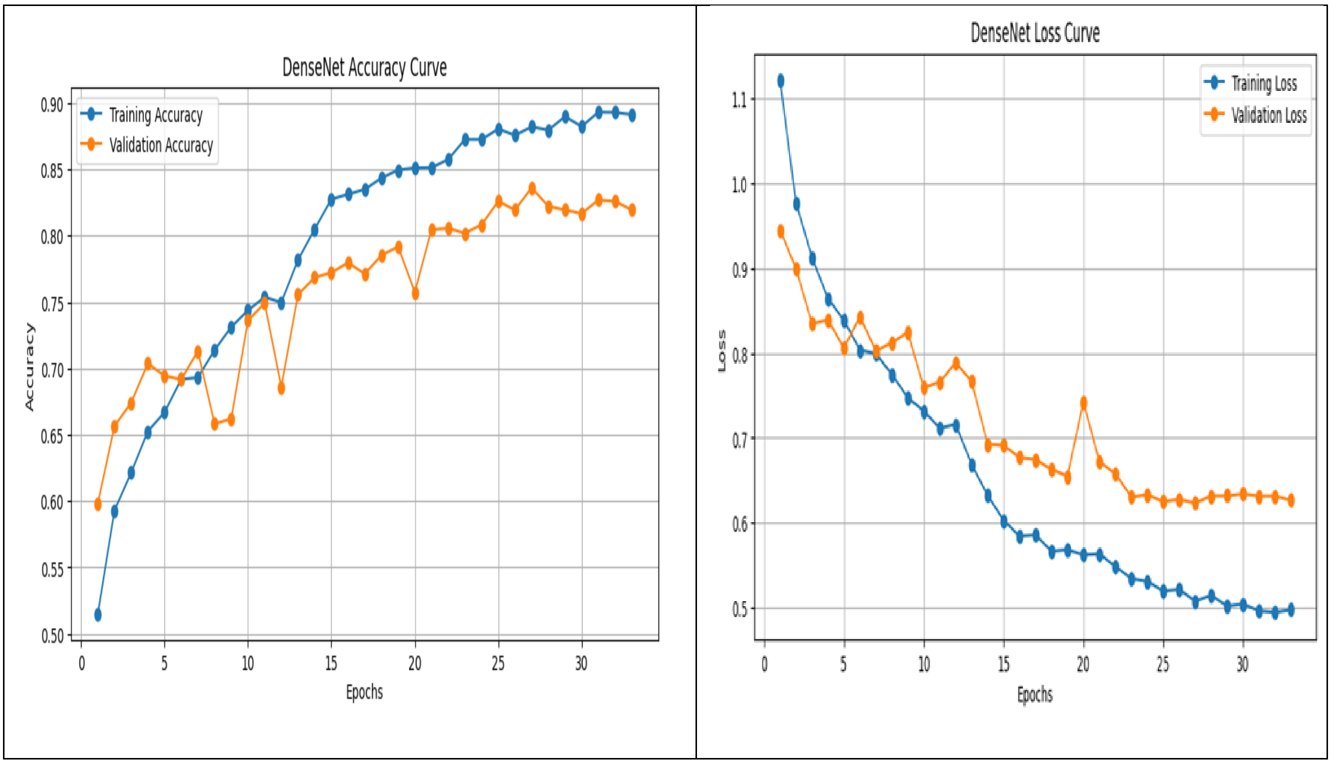


Fig 5.1 Accuracy and Loss Curves of DenseNet121 Model

Fig5.1 illustrates the DenseNet Accuracy and Loss Curves for both training and validation datasets across epochs.

The confusion matrix measures the DenseNet model's performance in classifying Normal, Osteopenia, and Osteoporosis classes by comparing actual and predicted labels for Knee Osteoarthritis detection from medical images. It provides a detailed visualization of correctly classified and misclassified samples for each category, helping evaluate the reliability of the model.

The matrix also helps identify class imbalance issues and regions where the model may confuse similar bone conditions during prediction. By analyzing false positives and false negatives, the confusion matrix supports performance improvement and enhances the overall accuracy and interpretability of the medical image classification system.

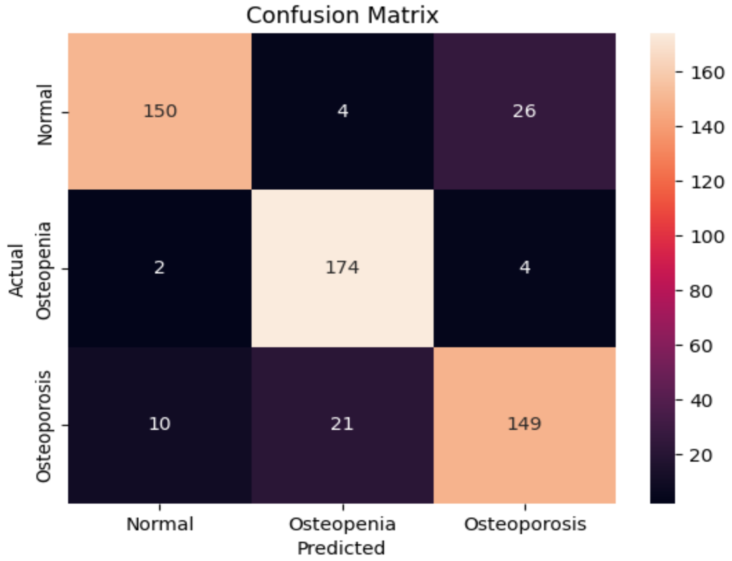


Fig 5.2 Confusion Matrix Analysis of DenseNet 121 Model

Fig 5.2 illustrates The graph compares the True Positive Rate (Sensitivity) against the False Positive Rate for different classification thresholds.

The ROC analysis confirms that the DenseNet model achieved strong classification accuracy and effective automated diagnosis of Normal, Osteopenia, and Osteoporosis conditions.

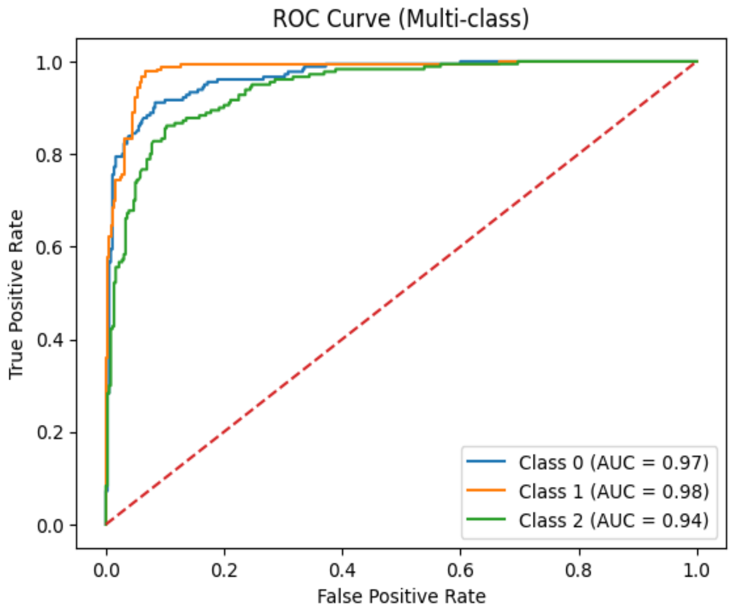


Fig 5.3 ROC Curve Analysis of DenseNet121 Model

Fig 5.3 iluustrates the ROC curve was used to evaluate the capability of the DenseNet architecture .

5.3.2 EfficientNet B3 Model

The accuracy and loss curves indicate stable model learning, with improving accuracy and decreasing prediction errors over epochs.

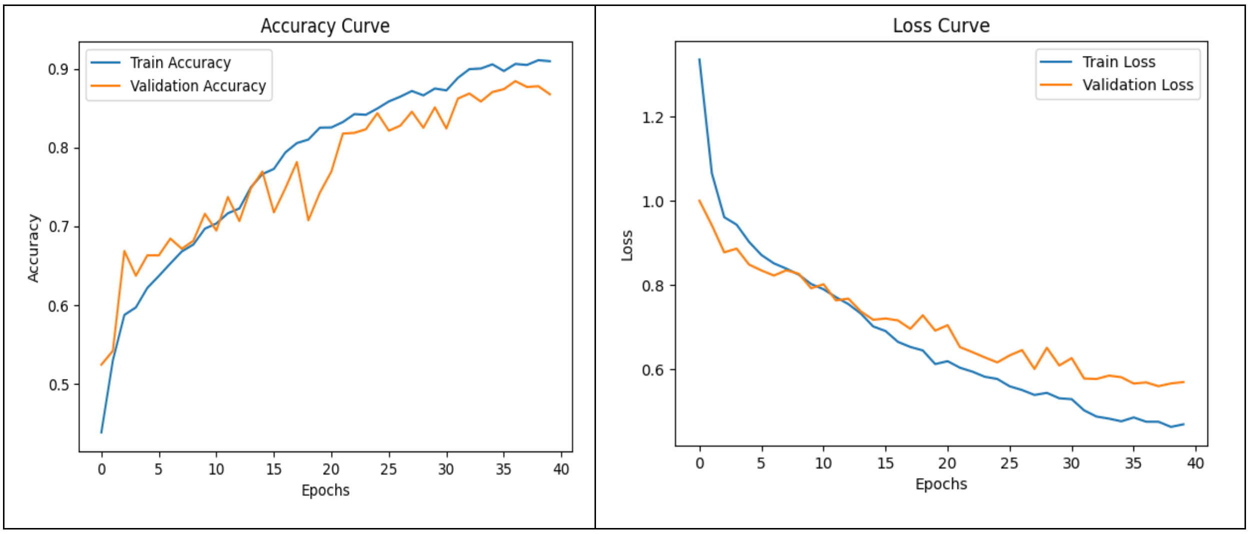


Fig 5.4 Accuracy and Loss Curves of EfficientNet B3 Model

Fig5.4 illustrates the training and validation performance of the deep learning model over 40 epochs of EfficientNet B3

The confusion matrix shows that the model correctly classified 165 Normal, 171 Osteopenia, and 163 Osteoporosis images, with only a few misclassifications between classes

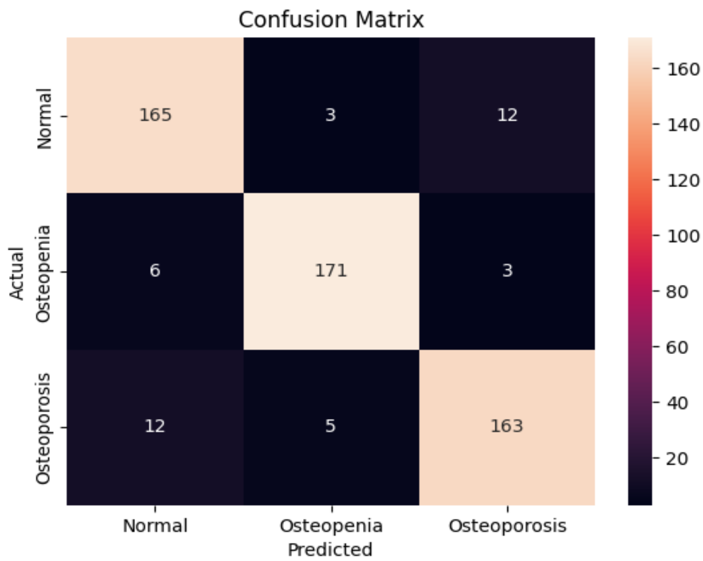


Fig 5.5 Confusion Matrix Analysis of EfficientNet B3 Model

Fig 5.5 illustrates the Confusion Matrix of EfficientNet B3 Model.

The ROC Curve evaluates classification performance by comparing sensitivity and false positive rate at different thresholds. A curve closer to the top-left corner with a high AUC value indicates excellent accuracy in distinguishing Normal, Osteopenia, and Osteoporosis classes.

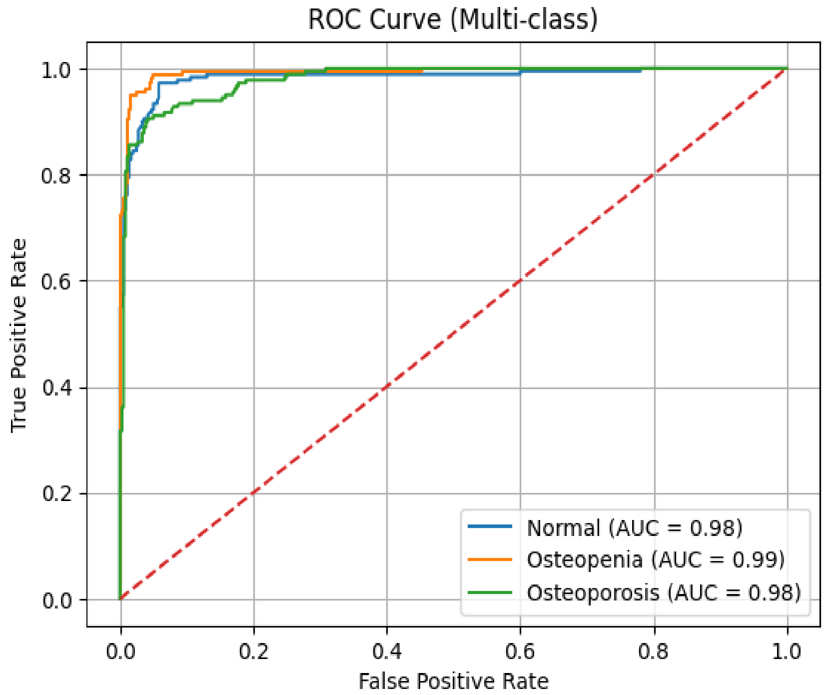


Fig 5.6 ROC Curve Analysis of EfficientNet B3 Model

Figure 5.6 represents the confusion matrix generated from the DenseNet model predictions for three classes Normal, Osteopenia, and Osteoporosis.

5.3.3 ResNet50V2 Model

The accuracy and loss graphs illustrate the learning performance of the ResNet50V2 model during training. The increasing accuracy and decreasing loss values across epochs indicate successful feature learning from Knee X-ray images, while the close relationship between training and validation curves demonstrates stable performance and good generalization capability

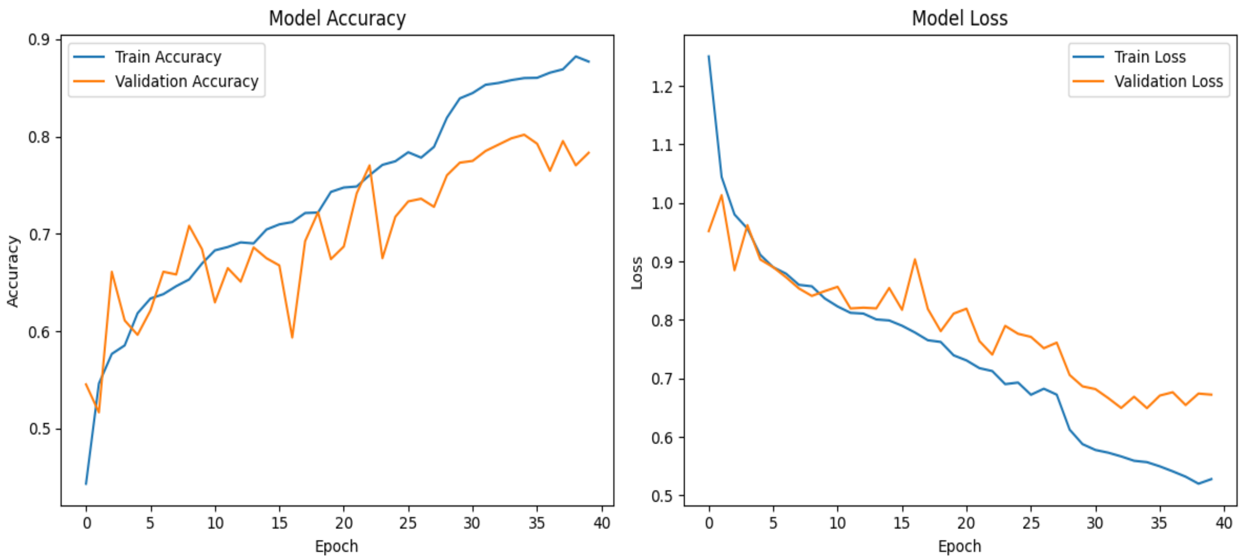


Fig 5.7 Accuracy and Loss Curves of ResNet50V2 Model

Fig 5.7 ResNet50V2 achieved stable convergence, high classification accuracy, and effective performance in detecting knee osteoarthritis from X-ray images..

The confusion matrix evaluates the model's classification performance by comparing actual and predicted class labels, where diagonal values indicate correct classifications and off-diagonal values represent misclassifications.

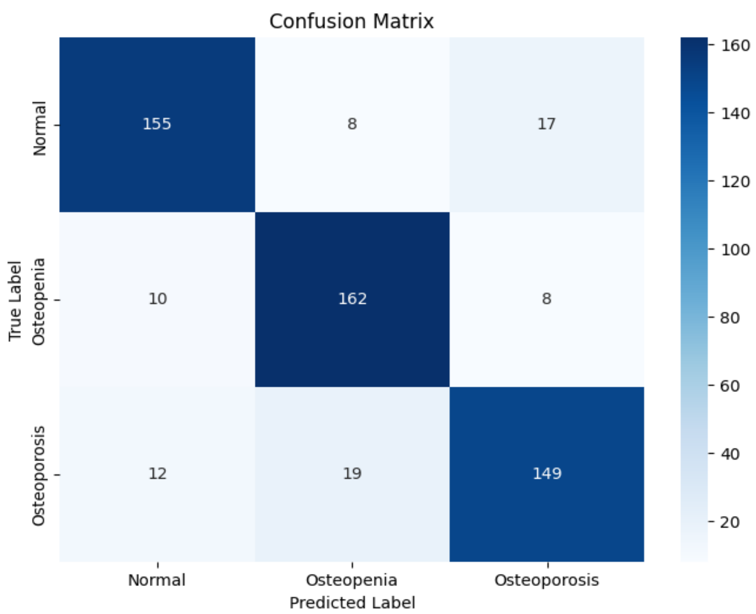


Fig 5.8 Confusion Matrix Analysis of ResNet50V2 Model

Fig 5.8 illustrates the confusion matrix of the ResNet50V2 model for three classes: Normal, Osteopenia, and Osteoporosis.

The multi-class ROC curve evaluates the model’s classification capability by comparing the True Positive Rate and False Positive Rate for each class. Higher AUC values indicate stronger prediction performance and better classification accuracy.

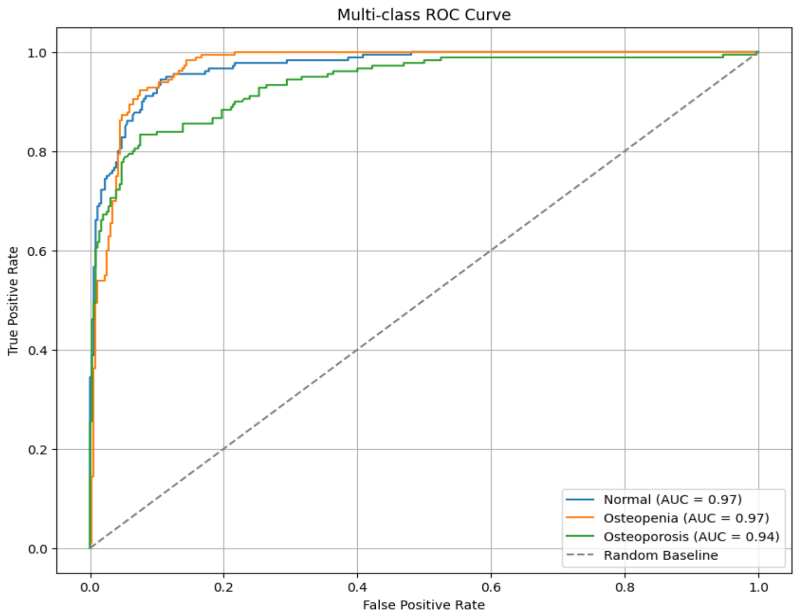


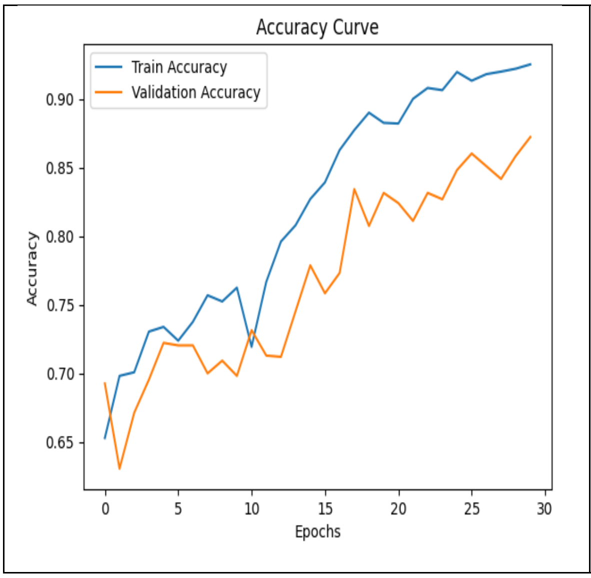
Fig 5.9 ROC Curve Analysis of ResNet50V2 Model

Fig5.9 ROC curve illustrates the classification performance of the ResNet50V2 model by comparing the True Positive Rate and False Positive Rate for multiple classes. The high AUC values indicate that the model achieves strong prediction accuracy and effective discrimination between normal and osteoarthritis classes.

5.3.4 Hybrid Model

The hybrid deep learning model combines EfficientNetV2-S and ConvNeXt-Tiny to extract local and global image features for accurate classification. It improves feature learning, reduces prediction errors, and achieves high accuracy for advanced image analysis tasks.

The hybrid model demonstrates strong classification capability by achieving high accuracy and low loss values, confirming efficient feature extraction and stable prediction performance on both training and validation datasets.



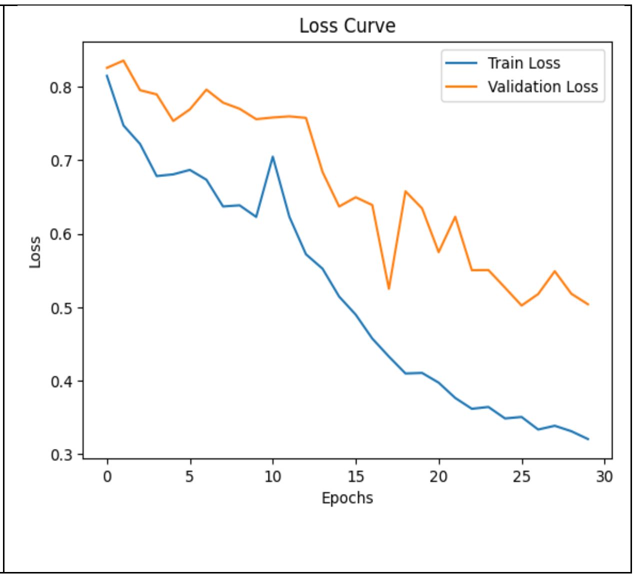


Fig 5.10 Accuracy and Loss Curves of Hybrid Model

Fig5.10 represents the Accuracy and Loss curves of the Hybrid Model during training and validation over multiple epochs.

The confusion matrix compares actual and predicted class labels, showing correct and incorrect classifications. Most values are concentrated along the diagonal, indicating high classification accuracy with minimal misclassification.

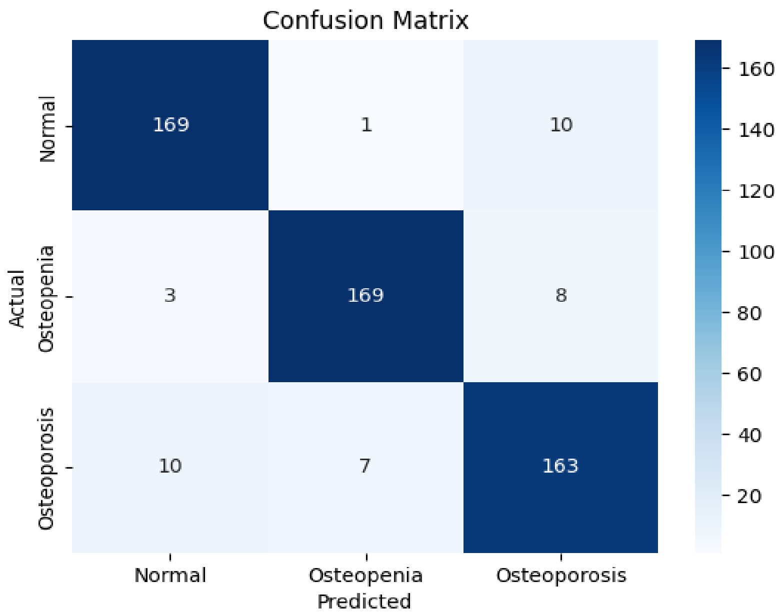


Fig 5.11 Confusion Matrix Analysis of Hybrid Model

Fig5.11 represents the Confusion Matrix of the hybrid model used for classifying Normal, Osteopenia, and Osteoporosis images.

The ROC analysis confirms that the proposed deep learning model achieves high classification accuracy and strong discriminative performance across all classes.

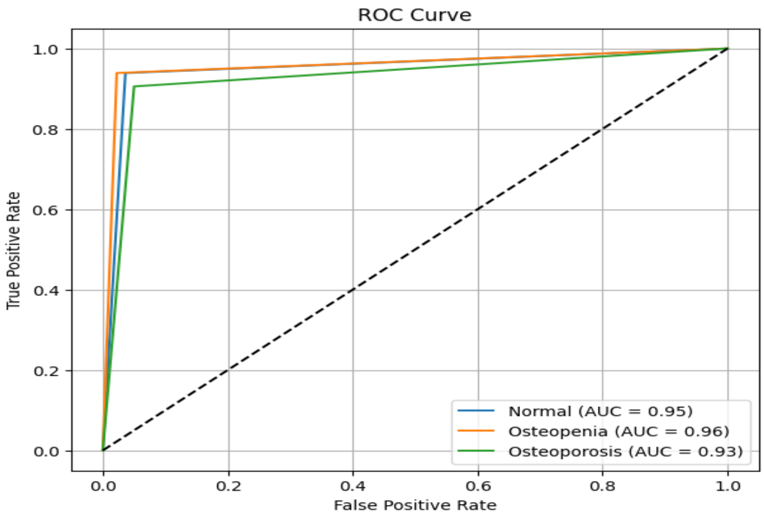


Fig5.12 ROC Curve Analysis of Hybrid Model

Fig 5.12 represents a multi-class ROC curve used to evaluate the classification performance of the deep learning model.

The Grad-CAM visualization confirms that the proposed deep learning model accurately focuses on clinically important regions of the Knee X-ray image, improving classification interpretability, reliability, and explainable AI performance in medical diagnosis.

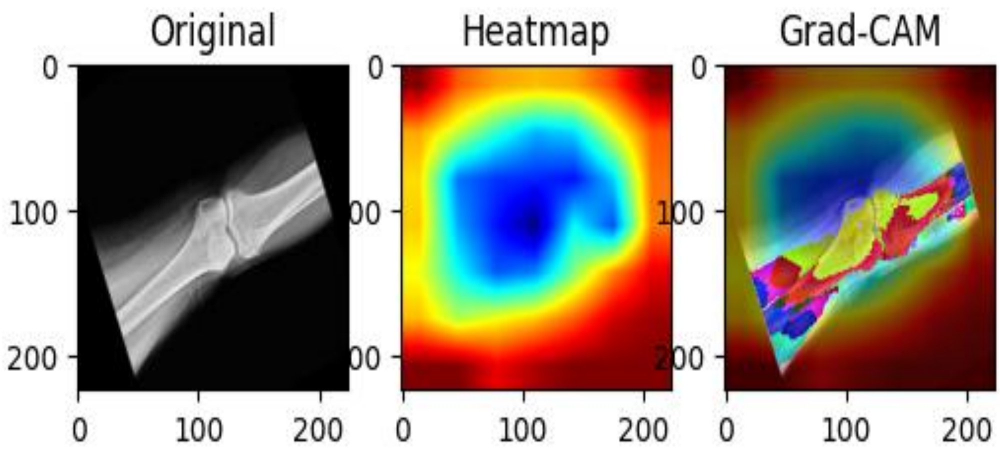


Fig 5.13 Grad-CAM Visualization of Knee DXA Image Classification

Fig5.13 represents a Grad-CAM visualization generated by the deep learning model to identify important regions in the Knee X-ray image used for classification

The Grad-CAM visualizations show that the deep learning model accurately focuses on important Knee X-ray regions for Normal class prediction, demonstrating reliable classification performance and effective explainable AI capability.

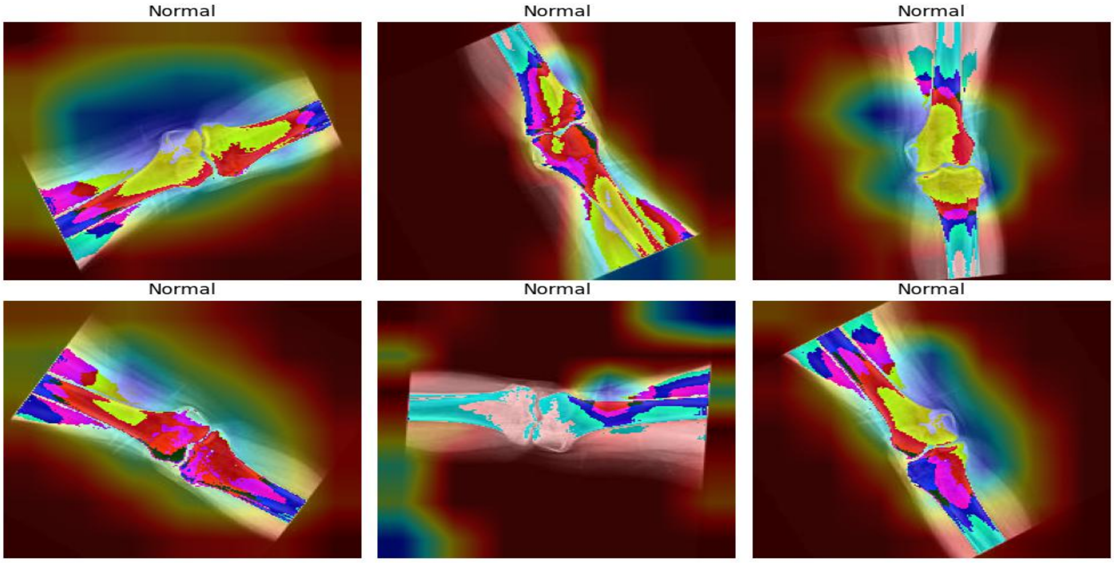


Fig5.14Grad-CAM Visualization for Normal Knee DXA Image Classification

Fig5.14 represents multiple Grad-CAM visualizations generated by the deep learning model for Normal Knee X-ray images.

SUMMARY

This chapter presents the results and discussion of deep learning models for Knee Osteoarthritis (KOA) classification using Knee X-ray images. Models including DenseNet121, EfficientNetB3, ResNet50V2, and a Hybrid architecture were trained using preprocessing techniques such as resizing, normalization, and data augmentation to improve performance and reduce overfitting. Performance was evaluated using metrics like accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC analysis. Among the models, EfficientNetB3 and the Hybrid model achieved strong classification accuracy and stable learning performance with minimal prediction errors. ROC curves and confusion matrices confirmed effective discrimination between Normal, Osteopenia, and Osteoporosis classes. Grad-CAM visualizations further demonstrated that the proposed models focused on clinically important Knee X-ray regions, improving interpretability and reliability in medical diagnosis.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

The classification of knee osteoarthritis using deep learning techniques demonstrated that advanced architectures and hybrid models significantly enhance diagnostic performance. Among the models implemented, EfficientNet, DenseNet, and hybrid approaches achieved high accuracy by effectively extracting both local and global features from X-ray images. Transfer learning improved performance on limited datasets, while techniques such as data augmentation and regularization helped prevent overfitting. The results confirm that combining multiple models and optimizing training strategies leads to a more robust and reliable system for automated medical image classification.

6.2 FUTURE ENHANCEMENT

Future improvements for the proposed Knee Osteoarthritis detection system include incorporating additional clinical information such as patient history, age, and symptoms to improve prediction accuracy and reliability. Advanced hyperparameter optimization methods like Bayesian optimization and Genetic Algorithms can further enhance model performance, while more sophisticated hybrid architectures combining CNNs, transformers, and detection models may improve classification accuracy. Expanding the dataset with advanced augmentation and synthetic data generation can strengthen model generalization. Additionally, integrating explainable AI techniques such as SHAP and LIME can improve interpretability and clinical trust, and developing web or mobile-based real-time deployment systems can increase practical usability in healthcare applications.

SUMMARY

The study demonstrates that deep learning models such as EfficientNet, DenseNet, ResNet, Swin Transformer, YOLO, and hybrid approaches effectively classify knee osteoarthritis from X-ray images with high accuracy and reliability.Techniques like transfer learning, data augmentation, regularization, and ensemble methods improved feature extraction.

APPENDIX

SAMPLE CODING

A.DENSENET121 MODEL

```
import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import DenseNet121
from tensorflow.keras.applications.densenet import preprocess_input
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Dense, Dropout, GlobalAveragePooling2D,
BatchNormalization
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
train_path = "/content/drive/MyDrive/Knee Osteoarthritis Classification/train"
val_path = "/content/drive/MyDrive/Knee Osteoarthritis Classification/val"
test_path = "/content/drive/MyDrive/Knee Osteoarthritis Classification/test"
IMG_SIZE = 224
BATCH_SIZE = 32
train_datagen = ImageDataGenerator(
    preprocessing_function=preprocess_input,
    rotation_range=15,
    zoom_range=0.15,
    horizontal_flip=True
)
val_datagen = ImageDataGenerator(preprocessing_function=preprocess_input)
train_data = train_datagen.flow_from_directory(
    train_path,
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    class_mode='categorical'
)
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
```

```
    class_mode='categorical'
)
test_data = val_datagen.flow_from_directory(
    test_path,
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    class_mode='categorical',
    shuffle=False
)
base_model = DenseNet121(
    weights='imagenet',
    include_top=False,
    input_shape=(224,224,3)
)
for layer in base_model.layers:
    layer.trainable = True
x = base_model.output
x = GlobalAveragePooling2D()(x)
x = BatchNormalization()(x)
x = Dense(512, activation='relu')(x)
x = Dropout(0.5)(x)
x = Dense(256, activation='relu')(x)
x = Dropout(0.4)(x)
    min_lr=1e-6
)
history = model.fit(
    train_data,
    validation_data=val_data,
    epochs=35,
    callbacks=[early_stop, reduce_lr]
)
```

```
model.save("/content/drive/MyDrive/densenet_model.keras")
test_loss, test_acc = model.evaluate(test_data)
print("DenseNet Test Accuracy:", test_acc)
```

B.EFFICIENTNET B3 MODEL

```
import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import EfficientNetB3
from tensorflow.keras.applications.efficientnet import preprocess_input
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Dense, Dropout, GlobalAveragePooling2D,
BatchNormalization
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
train_path = "/content/drive/MyDrive/Knee Osteoarthritis Classification/train"
val_path = "/content/drive/MyDrive/Knee Osteoarthritis Classification/val"
test_path = "/content/drive/MyDrive/Knee Osteoarthritis Classification/test"
IMG_SIZE = 300
BATCH_SIZE = 16
train_datagen = ImageDataGenerator(
    preprocessing_function=preprocess_input,
    rotation_range=15,
    zoom_range=0.15,
    width_shift_range=0.05,
    height_shift_range=0.05,
    horizontal_flip=True
)
val_datagen = ImageDataGenerator(preprocessing_function=preprocess_input)
train_data = train_datagen.flow_from_directory(
    train_path,
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    class_mode='categorical'
```

```
)
val_data = val_datagen.flow_from_directory(
    val_path,
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    class_mode='categorical'
)
test_data = val_datagen.flow_from_directory(
    test_path,
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    class_mode='categorical',
    shuffle=False
)
base_model = EfficientNetB3(
    weights='imagenet',
    include_top=False,
    input_shape=(IMG_SIZE, IMG_SIZE, 3)
)
for layer in base_model.layers:
    layer.trainable = True
x = base_model.output
x = GlobalAveragePooling2D()(x)
x = BatchNormalization()(x)
x = Dense(512, activation='relu')(x)
x = Dropout(0.5)(x)
x = Dense(256, activation='relu')(x)
x = Dropout(0.4)(x)
x = Dense(128, activation='relu')(x)
x = Dropout(0.3)(x)
output = Dense(3, activation='softmax')(x)
```

```
model = Model(inputs=base_model.input, outputs=output)
loss_fn = tf.keras.losses.CategoricalCrossentropy(label_smoothing=0.1)
model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-4),
    loss=loss_fn,
    metrics=['accuracy']
)
early_stop = EarlyStopping(
    monitor='val_loss',
    patience=6,
    restore_best_weights=True
)
reduce_lr = ReduceLROnPlateau(
    monitor='val_loss',
    factor=0.3,
    patience=3,
    min_lr=1e-6
)
history = model.fit(
    train_data,
    validation_data=val_data,
    epochs=40,
    callbacks=[early_stop, reduce_lr]
)
test_loss, test_acc = model.evaluate(test_data)
print("Final Test Accuracy:", test_acc)
```

C.RESTNET 50 V2

```
import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import ResNet50V2
from tensorflow.keras.applications.resnet_v2 import preprocess_input
```

```
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Dense, Dropout, GlobalAveragePooling2D, BatchNormalization
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau
train_path = "/content/drive/MyDrive/Colab Notebooks/KOA/Knee Osteoarthritis Classification/train"
val_path = "/content/drive/MyDrive/Colab Notebooks/KOA/Knee Osteoarthritis Classification/val"
test_path = "/content/drive/MyDrive/Colab Notebooks/KOA/Knee Osteoarthritis Classification/test"


IMG_SIZE = 300
BATCH_SIZE = 16
train_datagen = ImageDataGenerator(
    preprocessing_function=preprocess_input,
    rotation_range=15,
    zoom_range=0.15,
    width_shift_range=0.05,
    height_shift_range=0.05,
    horizontal_flip=True
)
val_datagen = ImageDataGenerator(preprocessing_function=preprocess_input)
train_data = train_datagen.flow_from_directory(
    train_path,
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    class_mode='categorical'
)
val_data = val_datagen.flow_from_directory(
    val_path,
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    class_mode='categorical'
```

```
)base_model = ResNet50V2( # Changed to ResNet50V2
    weights='imagenet',
    include_top=False,
    input_shape=(IMG_SIZE, IMG_SIZE, 3)
)
for layer in base_model.layers:
    layer.trainable = True
x = base_model.output
x = GlobalAveragePooling2D()(x)
x = BatchNormalization()(x)
x = Dense(512, activation='relu')(x)
x = Dropout(0.5)(x)
x = Dense(256, activation='relu')(x)
x = Dropout(0.4)(x)
x = Dense(128, activation='relu')(x)
x = Dropout(0.3)(x)
output = Dense(3, activation='softmax')(x)
model = Model(inputs=base_model.input, outputs=output)
history = model.fit(
    train_data,
    validation_data=val_data,
    epochs=40,
    callbacks=[early_stop, reduce_lr]
)
test_loss, test_acc = model.evaluate(test_data)
print("Final Test Accuracy:", test_acc)
```

D.HYBRID MODEL

```
import os
import torch
import timm
import numpy as np
```

```
import json
import matplotlib.pyplot as plt
import torch.nn as nn
import torch.optim as optim
from torchvision import datasets, transforms
from torch.utils.data import DataLoader, WeightedRandomSampler
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.utils.class_weight import compute_class_weight
from torch.amp import autocast, GradScaler
IMG_SIZE = 224
BATCH_SIZE = 16
EPOCHS = 30
PATIENCE = 6
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
train_transform = transforms.Compose([
    transforms.Resize((224,224)),
    transforms.RandomHorizontalFlip(),
    transforms.RandomRotation(8),
    transforms.ToTensor(),
    transforms.Normalize([0.485,0.456,0.406],
                         [0.229,0.224,0.225])
])
data_dir = "/content/drive/My Drive/Knee Osteoarthritis Classification"
train_dataset = datasets.ImageFolder(os.path.join(data_dir, "train"),
transform=train_transform)
val_dataset   = datasets.ImageFolder(os.path.join(data_dir, "val"),
transform=test_transform)
test_dataset  = datasets.ImageFolder(os.path.join(data_dir, "test"),
transform=test_transform)
targets = train_dataset.targets
class_sample_count = np.array(
    [len(np.where(np.array(targets) == t)[0]) for t in np.unique(targets)]
)
```

```
    def forward(self, x):
        return x * self.attn(x)
class HybridModel(nn.Module):
    def __init__(self, num_classes):
        super().__init__()
        self.eff = timm.create_model("tf_efficientnetv2_s", pretrained=True, num_classes=0)
        self.conv = timm.create_model("convnext_tiny", pretrained=True, num_classes=0)
            nn.ReLU(),
    if epoch == 5:
        print("Unfreezing backbone...")
        for p in model.parameters():
            p.requires_grad = True
        images, labels = images.to(device), labels.to(device)
        scaler.update()
        running_loss += loss.item()
        correct += (outputs.argmax(1) == labels).sum().item()
    train_loss = running_loss / len(train_loader)
    train_acc = correct / len(train_dataset)
    model.eval()
    val_loss = 0

    history = {
        "train_loss": train_losses,
        "val_loss": val_losses,
        "train_acc": train_acc_list,
        "val_acc": val_acc_list
    }
    preds = outputs.argmax(1)
    all_preds.extend(preds.cpu().numpy())
    all_labels.extend(labels.numpy())
print(classification_report(all_labels, all_preds, target_names=class_names))
```

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SUSTAINABLE DEVELOPMENT GOALS

Goal 3: Good Health and Well-Being

The proposed project primarily aligns with **SDG 3**, which focuses on ensuring healthy lives and promoting well-being for people of all ages. The Hybrid Deep Learning-Based Osteoporosis Classification System contributes to this goal by supporting early bone disease detection, intelligent medical image analysis, and improved healthcare decision-making0

.**1. Early Detection of Bone Disorders**

The system helps in the early identification of **Normal, Osteopenia, and Osteoporosis** conditions, enabling timely medical treatment and reducing the risk of severe bone complications.

2. Integration of Artificial Intelligence in Healthcare

The project combines EfficientNetV2-S and ConvNeXt-Tiny architectures with attention-based feature fusion to improve medical image classification performance. This integration of AI in healthcare supports the development of intelligent and automated diagnostic systems for bone health assessment.

3. Support for Clinical Decision-Making

The proposed model assists healthcare professionals by reducing manual diagnostic effort and improving classification accuracy through explainable AI techniques like **Grad-CAM**.

4. Promotion of Advanced and Scalable Healthcare Solutions

The project also promotes scalable and technology-driven healthcare systems that can support large-scale osteoporosis screening and clinical decision-making.

SIGNATURE OF THE SUPERVISOR